RANDOM VARIABLE

Random variable : Any real valued function which is defined on a sample space is known as a random variable.

(or)

A variable which takes values from a random experiment is known as a random variable.

Ex : If we toss THREE coins at a time then sample space S = { TTT, TTH, HTH, THH, HTT, THT, HHT, HHH }

If we assumed that " H " is a favorable case and the number of heads appeared is taken as a random variable ' x ' then it takes the values 0, 1, 2, 3

Types of Random variables : Random variables are classified into TWO types . They are

1. Discrete Random variable
2. Continuous Random variable

1. **Discrete Random Variable :** A random variable which takes finite number of values , then such variable is known as discrete random variable.

   EX : In rolling a die , a random variable ' x' takes 6 values. i. e., S= { 1, 2, 3, 4, 5, 6 }

2. **Continuous Random variable :** A random variable which takes all possible values between certain limits is known as continuous random variable.

(or)

If a random variable takes infinite number of values then such variable is known as continuous random variable.

Probability Function : Probability function can be defined in two types. They are

1. Probability mass function
2. Probability density function

1. **Probability Mass Function :** If ' x' is a discrete random variable then the probability function $P(x) = P(X = x) \ \forall \ x$ is called probability mass function. If and only if it satisfies
   - $P(x) \geq 0$
   - $0 \leq P(x) \leq 1$
   - $\sum_{x=0}^{n} P(x) = 1$
2. **Probability Density function :** If ' x' is a continuous random variable then the probability function $f(x)$ is given by $f(x) = P\left[x - \frac{dx}{2} < x < x + \frac{dx}{2}\right] \ \forall \ x$ is called probability density function and it must satisfies the following conditions.
   - $f(x) \geq 0$
   - $0 \leq f(x) \leq 1$
   - $\int_{-\infty}^{\infty} f(x)dx = 1$

Distribution Function : distribution functions are of TWO types. They are :

1. Discrete distribution function
2. Continuous distribution function

1. **Discrete distribution function :** If ' x' is a discrete random variable then the distribution function $F(x)$ is given by $F(x) = P(X = x) = \sum_{t=0}^{n} P(t)$ where $-\infty < x < \infty$

   Here , t is a value of ' x' which is less than or equal to ' x'.

   $F(x)$ is called discrete distribution function of a discrete random variable ' x '

2. **Continuous distribution function :** If ' x' is a continuous random variable then the continuous distribution function is given by $F(x) = \int_{-\infty}^{\infty} f(x)dx$ here F(x) is called continuous distribution function of a continuous random variable ' x'.

Properties of Distribution function :

1. If $F(x)$ is the distribution function of a random variable X and if $a < b$, then $P(a < X \le b) = F(b) - F(a)$
2. $P(a < X < b) = F(b) - F(a) - P(X = b)$
3. $P(a \le X \le b) = P(X = a) + F(b) - F(a)$
4. $P(a \le X < b) = F(b) - F(a) - P(X = b) + P(X = a)$
5. If $F(x)$ is the distribution function of a random variable X , then $0 \le F(x) \le 1$
6. If $F(x)$ is the distribution function of a random variable X ,then $F(x) \le F(y)$ for $x < y$.
7. If $F(x)$ is the distribution function of a random variable X, then $F(-\infty) = 0,\ F(\infty) = 1$

Various Formulae on Discrete Random Variable :

1. Mean = $\sum x.P(x)$
2. Variance = $\sum x^2 P(x) - (Mean)^2$
3. $F(x) = P(X \le x) = \sum^{x} P(X = x)$

Various Formulae on continuous Random Variable :

1. Mean= $\int x\ f(x)dx$
2. Variance = $\int x^2 f(x)dx - (Mean)^2$

Problem 1 : A random variable X has the following probability function.

x	-2	-1	0	1	2	3
$P(x)$	0.1	K	0.2	2K	0.3	K

Find (i) The value of K (ii) Mean and Variance (iii) Construct distribution function

Solution :

(i) By the definition of probability mass Function

$$\sum P(x) = 1$$

$$P(-2) + P(-1) + P(0) + P(1) + P(2) + P(3) = 1$$

$$0.1 + K + 0.2 + 2K + 0.3 + K = 1$$

$$0.6 + 4K = 1$$

$$4K = 1 - 0.6 = 0.4$$

$$\therefore \ K = 0.1$$

x	-2	-1	0	1	2	3
$P(x)$	0.1	0.1	0.2	0.2	0.3	0.1

(i)

Mean = $\sum x\,P(x)$

$= (-2).P(-2) + (-1)P(-1) + 0.P(0) + 1.P(1) + 2.P(2)$

$= (-2)(0.1) + (-1)(0.1) + 0 + 1.(0.2) + 2(0.3) + 3(0.1)$

$= -0.2 - 0.1 + 0.2 + 0.6 + 0.3$

$\therefore mean = 0.8$

Variance = $\sum x^2 P(x) - (Mean)^2$

$= (-2)^2.P(-2) + (-1)^2 P(-1) + 0^2.P(0) + 1^2.P(1) + 2^2.P(2) - (0.8)^2$

$= (-2)^2(0.1) + (-1)^2(0.1) + 0 + 1^2.(0.2) + 2^2(0.3) + 3^2(0.1) - (0.8)^2$

=n $0.4 + 0.1 + 0.2 + 1.2 + 0.9 - 0.64$

$\therefore Variance = 2.16$

(ii) Distribution function at the values of x can be calculated as follows

$$F(-2) = P(X \le -2) = P(-2) = 0.1$$

$$F(-1) = P(X \le -1) = P(-2) + P(-1) = 0.1 + 0.1 = 0.2$$

$$F(0) = P(X \le 0) = P(-2) + P(-1) + P(0) = 0.1 + 0.1 + 0.2 = 0.4$$

$$F(1) = P(X \le 1) = P(-2) + P(-1) + P(0) + P(1) = 0.1 + 0.1 + 0.2 + o.2 = 0.6$$

$$F(2) = P(X \le 2) = P(-2) + P(-1) + P(0) + P(1) + P(2) = 0.1 + 0.1 + 0.2 + o.2 + 0.3 = 0.9$$

$$F(3) = P(X \le 3) = P(-2) + P(-1) + P(0) + P(1) + P(2) + P(3) = 0.1 + 0.1 + 0.2 + o.2 + 0.3 + 0.1 = 1$$

Two dimensional random variables : Let x and y be any two random variables defined on a sample space ' S' then the function (x, y) that assigns a probability is called a two dimensional random variable.

Let (x, y) be a two dimensional random variable defined on a sample space 'S' and $\omega \in S$ then the value of x, y at ω is given by the pair of real numbers $\{X(\omega), Y(\omega)\}$

Let all the events $\omega \in S$ such that $X(\omega) \le a$ and $Y(\omega) \le b$ then the probability of an event is denoted as $P(x \le a, y \le b)$

Joint Probability Function : If (x, y) is a two dimensional discrete random variable then the joint discrete function of x, y is called As the joint probability mass function of (x, y) and it is denoted by P_{XY} and it is defined as $P_{XY}(x_i y_j) = P(X = x_i, Y = y_j)$.

If x and y are independent then $P_{XY}(x_i, y_j) = 0$

	Y_1	Y_2	Y_3	.....	Y_j	.....	Y_n	Total
X_1	P_{11}	P_{12}	P_{13}	.....	P_{1j}	.....	P_{1n}	$P_{1.}$
X_2	P_{21}	P_{22}	P_{23}	.....	P_{2j}	.....	P_{2n}	$P_{2.}$
X_3	P_{31}	P_{32}	P_{33}	.....	P_{3j}	.....	P_{3n}	$P_{3.}$
⋮	⋮	⋮	⋮	.....	⋮	.....	⋮	⋮
x_i	P_{i1}	P_{i2}	P_{i3}	.....	P_{ij}	.....	P_{in}	$P_{i.}$
⋮	⋮	⋮	⋮	.....	⋮	.....	⋮	⋮
X_m	P_{m1}	P_{m2}	P_{m3}	.....	P_{mj}	.....	P_{mn}	$P_{m.}$
Total	$P_{.1}$	$P_{.2}$	$P_{.3}$	.....	$P_{.j}$	.....	$P_{.n}$	1

Definition : The joint probability mass function of X and Y is denoted by $P(x_i, y_j)$ or p_{ij} or

$P(X = x_i, Y = y_j)$ and it should satisfy the following conditions.

i) $P(x_i, y_j) \ge 0 \forall (i, j)$

ii) $\sum_{i=1}^{m} \sum_{j=1}^{n} P(x_i, y_j) = 1$

1) The joint probability density function of X and Y is given by $f(x,y) = \frac{1}{8}(6 - x - y);\ 0 < x < 2,\ 2 < y < 4$

Find (i) $P(X < 1 \cap Y < 3)$ (ii) $P(X + Y < 3)$ (iii) Marginal probability density function of X and Y

(iii) $P\left(X < 1 / Y < 3\right)$

Sol : (1) $P(X<1\cap Y<3)=\int_0^1\int_2^3 f(x,y)dxdy$

$$=\int_0^1\int_2^3 \frac{1}{8}(6-x-y)dxdy$$

$$=\frac{1}{8}\int_0^1\left\{\int_2^3(6-x-y)dy\right\}dx$$

$$=\frac{1}{8}\int_0^1\left[6y-xy-\frac{y^2}{2}\right]_2^3 dx$$

$$=\frac{1}{8}\int_0^1\left[\left(18-3x-\frac{9}{2}\right)-\left(12-2x-\frac{4}{2}\right)\right]dx$$

$$=\frac{1}{8}\int_0^1\left(\frac{7}{2}-x\right)dx$$

$$=\frac{1}{8}\left[\frac{7}{2}x-\frac{x^2}{2}\right]_0^1$$

$$=\frac{1}{16}\left[7x-x^2\right]_0^1$$

$$=\frac{1}{16}\left[(7-1)-(0-0)\right]$$

$$=\frac{6}{16}$$

$$P(X<1\cap Y<3)=\frac{3}{8}$$

(2) $P(X+Y<3)=\int_0^1\int_2^{3-x} f(x,y)dxdy$

$$=\frac{1}{8}\int_0^1\left\{\int_2^{3-x}(6-x-y)dy\right\}dx$$

$$=\frac{1}{8}\int_0^1\left\{\left[6y-xy-\frac{y^2}{2}\right]_2^{3-x}\right\}dx$$

$$=\frac{1}{8}\int_0^1\left\{\left[6(3-x)-x(3-x)-\frac{(3-x)^2}{2}\right]-\left[6(2)-x(2)-\frac{2^2}{2}\right]\right\}dx$$

$$=\frac{1}{8}\int_0^1\left[\left(18-6x-3x+x^2-\frac{9-6x+x^2}{2}\right)-(12-2x-2)\right]dx$$

$$=\frac{1}{8}\int_0^1\left[\left(18-9x+x^2-\frac{9-6x+x^2}{2}\right)-(10-2x)\right]dx$$

$$= \frac{1}{8}\int_0^1 \left[\frac{\left(36 - 18x + 2x^2 - 9 + 6x - x^2 - 20 + 4x\right)}{2} \right] dx$$

$$= \frac{1}{8}\int_0^1 \left[\frac{\left(7 - 8x + x^2\right)}{2} \right] dx$$

$$= \frac{1}{16}\int_0^1 \left(x^2 - 8x + 7\right) dx$$

$$= \frac{1}{16}\left[\left(\frac{x^3}{3} - 8\frac{x^2}{2} + 7x \right)_0^1 \right]$$

$$= \frac{1}{16}\left[\left(\frac{1^3}{3} - 4\left(1^2\right) + 7(1) \right) - \left(\frac{0^3}{3} - 4(0) + 7(0) \right) \right]$$

$$= \frac{1}{16}\left[\left(\frac{1}{3} - 4 + 7 \right) - 0 \right]$$

$$= \frac{1}{16}\left[\frac{1}{3} + 3 \right] = \frac{1}{16}\left(\frac{10}{3} \right)$$

$$P\left(X + Y < 3\right) = \frac{5}{24}$$

(3) Marginal probability density function of X is

$$f(x) = \int_y f(x, y) dy$$

$$= \int_2^4 \frac{1}{8}(6 - x - y) dy$$

$$= \frac{1}{8}\int_2^4 (6 - x - y) dy$$

$$= \frac{1}{8}\left[6y - xy - \frac{y^2}{2} \right]_2^4$$

$$= \frac{1}{8}\left\{ \left[6(4) - x(4) - \frac{4^2}{2} \right] - \left[6(2) - x(2) - \frac{2^2}{2} \right] \right\}$$

$$= \frac{1}{8}\left[(24-4x-8)-(12-2x-2)\right]$$

$$= \frac{1}{8}\left[6-2x\right]$$

$$f(x) = \frac{3-x}{4}\ ;\ 0 < x < 2$$

Marginal probability density function of Y is

$$f(y) = \int_x f(x,y)dx$$

$$= \int_0^2 \frac{1}{8}(6-x-y)dx$$

$$= \frac{1}{8}\int_0^2 (6-x-y)dx$$

$$= \frac{1}{8}\left[6x - \frac{x^2}{2} - xy\right]_0^2$$

$$= \frac{1}{8}\left\{\left[6(2) - \frac{2^2}{2} - y(2)\right] - \left[6(0) - \frac{0^2}{2} - y(0)\right]\right\}$$

$$= \frac{1}{8}\left[(12-2-2y)-(0)\right]$$

$$= \frac{1}{8}\left[10-2y\right]$$

$$f(x) = \frac{5-y}{4}\ ;\ 2 < y < 4$$

(4) $$P(Y<3) = \int_2^3 f(y)dy$$

$$= \int_2^3 \frac{5-y}{4}dy$$

$$= \frac{1}{4}\int_2^3 (5-y)dy$$

$$= \frac{1}{4}\left[5y - \frac{y^2}{2}\right]_2^3$$

$$= \frac{1}{4}\left\{\left[5(3) - \frac{3^2}{2}\right] - \left[5(2) - \frac{2^2}{2}\right]\right\}$$

$$= \frac{1}{4}\left\{\left[15 - \frac{9}{2}\right] - \left[10 - \frac{4}{2}\right]\right\}$$

$$= \frac{1}{4}\left\{\left[15 - 10 - \frac{9}{2} + \frac{4}{2}\right]\right\} = \frac{1}{4}\left[5 - \frac{5}{2}\right] = \frac{1}{4}\left(\frac{5}{2}\right)$$

$$P(Y < 3) = \frac{5}{8}$$

$$\therefore\ P\left(X < 1 / Y < 3\right) = \frac{P\left(X < 1 \cap Y < 3\right)}{P\left(Y < 3\right)} = \frac{3/8}{5/8} = \frac{3}{5}$$

Discrete Distributions

Introduction : Distributions are broadly classified into two types. They are

i. Discrete Distributions
ii. Continuous Distributions

Binomial Distribution

Introduction :

Binomial distribution was introduced by Prof. James Bernoulli in the year 1700. This experiment can be used to study the distribution of success and failures in 'n' independent trails belongs to a random experiment.

If we consider 'n' independent Bernoulli trails with the probability of success 'p' and probability of failure 'q' remains constant then the probability of getting 'x' success out of 'n' trails is given by

$$P(X=x)=\begin{cases} {}^{n}C_{x}\,p^{x}q^{n-x} & ;\ x=0,1,2,3,.....n \\ 0 & ;\text{Otherwise} \end{cases}$$

Definition :

A random variable 'x' is said to follow Binomial distribution with parameter 'n' and 'p' if it assumes only non- negative values and its probability mass function is given by

$$P(X=x)=\begin{cases} {}^{n}C_{x}\,p^{x}q^{n-x} & ;\ x=0,1,2,3,.....n \\ 0 & ;\text{Otherwise} \end{cases}$$

Mean of a Binomial distribution :

By the definition of mathematical expectation, mathematical expectation of a random variable is equal to its arithmetic mean.

$\therefore$ Mean $= E(x)$

$$= \sum_{x} x\,P(X=x)$$

$$= \sum_{x=0}^{n} x\ {}^{n}C_{x}\,p^{x}q^{n-x}$$

$$= \sum_{x=0}^{n} x\,\frac{n!}{x!(n-x)!}\,p^{x}q^{n-x}$$

$$= \sum_{x=0}^{n} \cancel{x}\,\frac{n(n-1)!}{\cancel{x}(x-1)!(n-x)!}\,p\ p^{x-1}q^{n-x}$$

$$= n\,p\sum_{x=1}^{n} \frac{(n-1)!}{(x-1)!((n-1)-(x-1))!}\,p^{x-1}q^{n-x}$$

$$= n\,p\sum_{x=1}^{n} {}^{(n-1)}C_{x-1}\ p^{x-1}q^{n-x}$$

$$= n\,p\left\{{}^{(n-1)}C_{1-1}p^{1-1}q^{n-1}+{}^{(n-1)}C_{2-1}p^{2-1}q^{n-2}+{}^{(n-1)}C_{3-1}p^{3-1}q^{n-3}+{}^{(n-1)}C_{4-1}p^{4-1}q^{n-4}+.....+{}^{(n-1)}C_{n-1}p^{n-1}q^{n-n}\right\}$$

$$= n\,p\left\{{}^{(n-1)}C_0\,p^0 q^{n-1} + {}^{(n-1)}C_1\,p^1 q^{n-2} + {}^{(n-1)}C_2\,p^2 q^{n-3} + {}^{(n-1)}C_3\,p^3 q^{n-4} + \ldots\ldots + {}^{(n-1)}C_{n-1}\,p^{n-1} q^0\right\}$$

$$\boxed{(x+y)^n = {}^nC_0 x^0 y^n + {}^nC_1 x^1 y^{n-1} + {}^nC_2 x^2 y^{n-2} + {}^nC_3 x^3 y^{n-3} + {}^nC_4 x^4 y^{n-4} + \ldots.. + {}^nC_n x^n y^0}$$

$$Mean = n\,p\left[(p+q)^{n-1}\right] = n\,p\,[1]^{n-1} = n\,p(1) = np$$

$\therefore$ Mean of a Binomial distribution is 'np'

Consider $E(x^2) = \sum_x x^2\,P(X=x)$

$$= \sum_x (x^2 - x + x)P(X=x) = \sum_x (x^2 - x)P(X=x) + \sum_x x\,P(X=x)$$

$$= \sum_{x=0}^{n} x(x-1)\;{}^nC_x\,p^x q^{n-x} + np$$

$$= \sum_{x=0}^{n} x(x-1)\frac{n!}{x!(n-x)!}\,p^x q^{n-x} + np$$

$$= \sum_{x=0}^{n} \cancel{x(x-1)}\frac{n(n-1)(n-2)!}{\cancel{x(x-1)}(x-2)!(n-x)!}\,p^2\;p^{x-2} q^{n-x} + np$$

$$= n(n-1)\,p^2 \sum_{x=2}^{n} \frac{(n-2)!}{(x-2)!((n-2)-(x-2))!}\,p^{x-2} q^{n-x} + np$$

$$= n(n-1)\,p^2 \sum_{x=2}^{n} {}^{(n-2)}C_{x-2}\;p^{x-2} q^{n-x} + np$$

$$= (n^2 - n)p^2 \left\{ \begin{array}{l} {}^{(n-2)}C_{2-2}\,p^{2-2} q^{n-2} + {}^{(n-2)}C_{3-2}\,p^{3-2} q^{n-3} + {}^{(n-2)}C_{4-2}\,p^{4-2} q^{n-4} \\ \qquad\qquad + {}^{(n-2)}C_{5-2}\,p^{5-2} q^{n-5} + \ldots\ldots + {}^{(n-2)}C_{n-2}\,p^{n-2} q^{n-n} \end{array} \right\} + np$$

$$= (n^2 - n)\,p^2 \left\{{}^{(n-2)}C_0\,p^0 q^{n-2} + {}^{(n-2)}C_1\,p^1 q^{n-3} + {}^{(n-2)}C_2\,p^2 q^{n-4} + {}^{(n-2)}C_3\,p^3 q^{n-5} + \ldots\ldots + {}^{(n-2)}C_{n-2}\,p^{n-2} q^0\right\} + np$$

$$\boxed{(x+y)^n = {}^nC_0 x^0 y^n + {}^nC_1 x^1 y^{n-1} + {}^nC_2 x^2 y^{n-2} + {}^nC_3 x^3 y^{n-3} + {}^nC_4 x^4 y^{n-4} + \ldots.. + {}^nC_n x^n y^0}$$

$$E(x^2) = (n^2 p^2 - np^2)\left[(p+q)^{n-2}\right] + np$$

$$= (n^2 p^2 - np^2)[1]^{n-2} + np$$

$$= (n^2 p^2 - np^2)(1) + np$$

$$E(x^2) = n^2 p^2 - np^2 + np$$

By the definition of variance,

$$V(x) = E(x^2) - [E(x)]^2 = \left[n^2 p^2 - np^2 + np\right] - (np)^2 = \cancel{n^2 p^2} - np^2 + np - \cancel{n^2 p^2}$$

$$V(x) = np - np^2 = np(1-p) = npq$$

$\therefore$ Variance of a Binomial distribution is $V(x) = npq$

Moment generating function of a Binomial distribution :

A random variable 'x' is said to follow Binomial distribution with parameter 'n' and 'p' if it assumes only non- negative values and its probability mass function is given by

$$P(X=x)=\begin{cases} {}^nC_x p^x q^{n-x} & ;\ x=0,1,2,3,.....n \\ 0 & ;\text{Otherwise} \end{cases}$$

By the definition of moment generating function, moment generating function of a Binomial distribution is

$$M_X(t)=E\left[e^{tx}\right]$$

$$=\sum_x e^{tx}P(X=x)$$

$$=\sum_{x=0}^{n} e^{tx}\ {}^nC_x\ p^x\ q^{n-x}$$

$$=\sum_{x=0}^{n} {}^nC_x\left(e^t\right)^x\ p^x\ q^{n-x}$$

$$=\sum_{x=0}^{n} {}^nC_x\left(pe^t\right)^x\ q^{n-x}$$

$$={}^nC_0\left(pe^t\right)^0 q^{n-0}+{}^nC_1\left(pe^t\right)^1 q^{n-1}+{}^nC_2\left(pe^t\right)^2 q^{n-2}+{}^nC_3\left(pe^t\right)^3 q^{n-3}+.....+{}^nC_n\left(pe^t\right)^n q^{n-n}$$

$$\boxed{(x+y)^n={}^nC_0x^0y^n+{}^nC_1x^1y^{n-1}+{}^nC_2x^2y^{n-2}+{}^nC_3x^3y^{n-3}+{}^nC_4x^4y^{n-4}+.....+{}^nC_nx^ny^0}$$

$$M_X(t)=\left(pe^t+q\right)^n$$

$$M_X(t)=\left(q+pe^t\right)^n$$

$\therefore$ Moment generating function of a Binomial distribution is $M_X(t)=\left(q+pe^t\right)^n$

Characteristic function of a Binomial distribution :

A random variable 'x' is said to follow Binomial distribution with parameter 'n' and 'p' if it assumes only non- negative values and its probability mass function is given by

$$P(X=x)=\begin{cases} {}^nC_x p^x q^{n-x} & ;\ x=0,1,2,3,....n \\ 0 & ;\text{Otherwise} \end{cases}$$

By the definition of Characteristic function, moment generating function of a Binomial distribution is

$$\phi_X(t)=E\left[e^{itx}\right]$$

$$=\sum_x e^{itx}P(X=x)$$

$$=\sum_{x=0}^{n} e^{itx}\ {}^nC_x\ p^x\ q^{n-x}$$

$$=\sum_{x=0}^{n} {}^nC_x\left(e^{it}\right)^x\ p^x\ q^{n-x}$$

$$= \sum_{x=0}^{n} {}^{n}C_{x} \left(pe^{it}\right)^{x} q^{n-x}$$

$$= {}^{n}C_{0}\left(pe^{it}\right)^{0} q^{n-0} + {}^{n}C_{1}\left(pe^{it}\right)^{1} q^{n-1} + {}^{n}C_{2}\left(pe^{it}\right)^{2} q^{n-2} + {}^{n}C_{3}\left(pe^{it}\right)^{3} q^{n-3} + \ldots\ldots + {}^{n}C_{n}\left(pe^{it}\right)^{n} q^{n-n}$$

$$\boxed{(x+y)^{n} = {}^{n}C_{0}x^{0}y^{n} + {}^{n}C_{1}x^{1}y^{n-1} + {}^{n}C_{2}x^{2}y^{n-2} + {}^{n}C_{3}x^{3}y^{n-3} + {}^{n}C_{4}x^{4}y^{n-4} + \ldots + {}^{n}C_{n}x^{n}y^{0}}$$

$$\phi_{X}(t) = \left(pe^{it} + q\right)^{n}$$

$$\phi_{X}(t) = \left(q + pe^{it}\right)^{n}$$

$\therefore$ Characteristic function of a Binomial distribution is $\phi_{X}(t) = \left(q + pe^{it}\right)^{n}$

Moments of Binomial distribution :

Moment generating function of a Binomial distribution is given by $M_{X}(t) = \left(q + pe^{t}\right)^{n} \rightarrow (1)$

Partially differentiating equation (1) with respect to 't'

$$\frac{\partial}{\partial t} M_{X}(t) = \frac{\partial}{\partial t}\left(q + pe^{t}\right)^{n}$$

$$= n\left(q + pe^{t}\right)^{n-1}\left(0 + pe^{t}\right)$$

$$= n\left(q + pe^{t}\right)^{n-1}\left(pe^{t}\right)$$

$$\frac{\partial}{\partial t} M_{X}(t) = n\, p\, e^{t}\left(q + pe^{t}\right)^{n-1} \rightarrow (2)$$

We know that , $\mu_{1}' = \left.\frac{\partial}{\partial t} M_{X}(t)\right]_{t=0}$

$$= n\, p\, e^{0}\left(q + pe^{0}\right)^{n-1}$$

$$= n\, p\,(1)(q + p)^{n-1}$$

$$= n\, p\,(1)^{n-1}$$

$$\mu_{1}' = np$$

$\therefore$ First Raw moment of a Binomial distribution is $\mu_{1}' = np$

Partially differentiating equation (2) with respect to 't'

$$\frac{\partial^{2}}{\partial t^{2}} M_{X}(t) = \frac{\partial}{\partial t}\left\{n\, p\, e^{t}\left(q + pe^{t}\right)^{n-1}\right\}$$

$$= n\, p \frac{\partial}{\partial t}\left[e^{t}\left(q + pe^{t}\right)^{n-1}\right]$$

$$= np\left\{e^{t}.\left(q + pe^{t}\right)^{n-1} + e^{t}(n-1)\left(q + pe^{t}\right)^{n-2}\left(pe^{t}\right)\right\}$$

$$\frac{\partial^2}{\partial t^2} M_X(t) = np\left\{e^t.\left(q+pe^t\right)^{n-1} + (n-1)p\, e^{2t}\left(q+pe^t\right)^{n-2}\right\} \rightarrow (3)$$

We know that , $\mu_2' = \left.\frac{\partial^2}{\partial t^2} M_X(t)\right]_{t=0}$

$$\mu_2' = np\left\{e^0.\left(q+pe^0\right)^{n-1} + (n-1)p\, e^{2(0)}\left(q+pe^0\right)^{n-2}\right\}$$

$$\mu_2' = np\left\{1.(q+p)^{n-1} + (n-1)p\,.1.(q+p)^{n-2}\right\}$$

$$\mu_2' = np\left\{(1)^{n-1} + (n-1)p(1)^{n-2}\right\}$$

$$\mu_2' = np\{1+(n-1)p\} = np\{1+np-p\}$$

$$\mu_2' = n^2p^2 - np^2 + np$$

$\therefore$ Second Raw moment of a Binomial distribution is $\mu_2' = n^2p^2 - np^2 + np$

Partially differentiating equation (3) with respect to 't'

$$\frac{\partial^3}{\partial t^3} M_X(t) = \frac{\partial}{\partial t} np\left\{e^t.\left(q+pe^t\right)^{n-1} + (n-1)p\, e^{2t}\left(q+pe^t\right)^{n-2}\right\}$$

$$= np\left\{\frac{\partial}{\partial t}\left[e^t.\left(q+pe^t\right)^{n-1}\right] + (n-1)p\frac{\partial}{\partial t}\left[e^{2t}\left(q+pe^t\right)^{n-2}\right]\right\}$$

$$= np\left\{\begin{array}{l}\left[e^t.\left(q+pe^t\right)^{n-1} + e^t.(n-1)\left(q+pe^t\right)^{n-2}\left(0+pe^t\right)\right] \\ \qquad + (n-1)p\left[2e^{2t}\left(q+pe^t\right)^{n-2} + e^{2t}(n-2)\left(q+pe^t\right)^{n-3}\left(0+pe^t\right)\right]\end{array}\right\}$$

$$= np\left\{\begin{array}{l}\left[e^t.\left(q+pe^t\right)^{n-1} + (n-1)pe^{2t}.\left(q+pe^t\right)^{n-2}\right] \\ \qquad + (n-1)p\left[2e^{2t}\left(q+pe^t\right)^{n-2} + (n-2)pe^{3t}\left(q+pe^t\right)^{n-3}\right]\end{array}\right\}$$

$$= np\left\{\begin{array}{l}\left[e^t.\left(q+pe^t\right)^{n-1} + (n-1)pe^{2t}.\left(q+pe^t\right)^{n-2}\right] \\ \qquad + 2(n-1)pe^{2t}\left(q+pe^t\right)^{n-2} + (n-1)(n-2)p^2e^{3t}\left(q+pe^t\right)^{n-3}\end{array}\right\}$$

$$\frac{\partial^3}{\partial t^3} M_X(t) = np\left\{e^t.\left(q+pe^t\right)^{n-1} + 3(n-1)pe^{2t}.\left(q+pe^t\right)^{n-2} + (n-1)(n-2)p^2e^{3t}\left(q+pe^t\right)^{n-3}\right\} \rightarrow (4)$$

We know that $\mu_3' = \left.\frac{\partial^3}{\partial t^3} M_X(t)\right]_{t=0}$

$$\mu_3' = np\left\{e^0.\left(q+pe^0\right)^{n-1} + 3(n-1)pe^{2(0)}.\left(q+pe^0\right)^{n-2} + (n-1)(n-2)p^2e^{3(0)}\left(q+pe^0\right)^{n-3}\right\}$$

$$\mu_3' = np\{1.(q+p)^{n-1} + 3(n-1)p.1.(q+p)^{n-2} + (n-1)(n-2)p^2.1(q+p)^{n-3}\}$$

$$\mu_3' = np\{(1)^{n-1} + 3(n-1)p(1)^{n-2} + (n-1)(n-2)p^2(1)^{n-3}\}$$

$$\mu_3' = np\{1 + 3(n-1)p + (n-1)(n-2)p^2\}$$

$\therefore$ Third Raw moment of a Binomial distribution is $\mu_3' = np\{1 + 3(n-1)p + (n-1)(n-2)p^2\}$

Similarly,

$\therefore$ Fourth Raw moment of a Binomial distribution is

$$\mu_4' = np\{1 + 7(n-1)p + 6(n-1)(n-2)p^2 + (n-1)(n-2)(n-3)p^3\}$$

$$\mu_4' = np + 7n(n-1)p^2 + 6n(n-1)(n-2)p^3 + n(n-1)(n-2)(n-3)p^4$$

Central moments of Binomial Distribution :

By the relationship between Central and non central moments

$\therefore$ First central moment of a Binomial distribution is $\mu_1 = 0$

Second central moment $\mu_2 = \mu_2' - (\mu_1')^2$

$$\mu_2 = [n^2p^2 - np^2 + np] - (np)^2$$

$$\mu_2 = \cancel{n^2p^2} - np^2 + np - \cancel{n^2p^2}$$

$$\mu_2 = np(1-p) = npq$$

$\therefore$ Second Central moment of a Binomial distribution is $\mu_2 = npq$

Third central moment $\mu_3 = \mu_3' - 3\mu_2'(\mu_1') + 2(\mu_1')^3$

$$\mu_3 = [np + 3n(n-1)p^2 + n(n-1)(n-2)p^3] - 3[n^2p^2 - np^2 + np](np) + 2(np)^3$$

$$\mu_3 = np + \cancel{3n^2p^2} - 3np^2 + \cancel{n^3p^3} - \cancel{3n^2p^3} + 2np^3 - \cancel{3n^3p^3} + \cancel{3n^2p^3} - \cancel{3n^2p^2} + \cancel{2n^3p^3}$$

$$\mu_3 = np - 3np^2 + 2np^3$$

$$\mu_3 = np[2p^2 - 3p + 1]$$

$$\mu_3 = np[2p^2 - 2p - p + 1] \Rightarrow \mu_3 = np[2p(p-1) - 1(p-1)]$$

$$\mu_3 = np[(2p-1)(p-1)] \Rightarrow \mu_3 = np[(1-p)(1-2p)]$$

$$\mu_3 = npq[(p+q) - 2p] \Rightarrow \mu_3 = npq[q-p]$$

$\therefore$ Third Central moment of a Binomial distribution is $\mu_3 = npq[q-p]$

Similarly fourth central moment of a Binomial distribution is $\mu_4 = 3n^2p^2q^2 + npq[1-6pq]$

Coefficient of Skewness and Kurtosis of a Binomial Distribution :

Consider, $\beta_1 = \dfrac{\mu_3^2}{\mu_2^3} = \dfrac{\{npq[q-p]\}^2}{(npq)^3} = \dfrac{\cancel{(npq)^2}[q-p]^2}{(npq)^{\cancel{3}}} = \dfrac{(q-p)^2}{npq}$

We know that, $\gamma_1 = \sqrt{\beta_1} = \sqrt{\frac{(q-p)^2}{npq}} = \frac{q-p}{\sqrt{npq}}$

Coefficient of skewness of a Binomial distribution is $\gamma_1 == \frac{q-p}{\sqrt{npq}}$

Consider, $\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{3n^2p^2q^2 + npq[1-6pq]}{(npq)^2} = \frac{\cancel{3n^2p^2q^2}}{\cancel{n^2p^2q^2}} + \frac{\cancel{npq}[1-6pq]}{(npq)^{\cancel{2}}} = 3 + \frac{1-6pq}{npq}$

We know that, $\gamma_2 = \beta_2 - 3 = \cancel{3} + \frac{1-6pq}{npq} \cancel{-3} = \frac{1-6pq}{npq}$

Coefficient of kurtosis of a Binomial distribution is $\gamma_2 = \frac{1-6pq}{npq}$

Mean and Variance of a Binomial distribution using moment generating function :

Moment generating function of a Binomial distribution is given by $M_X(t) = (q + pe^t)^n \rightarrow (1)$

Partially differentiating equation (1) with respect to 't'

$$\frac{\partial}{\partial t} M_X(t) = \frac{\partial}{\partial t}(q + pe^t)^n$$

$$= n(q + pe^t)^{n-1}(0 + pe^t)$$

$$= n(q + pe^t)^{n-1}(pe^t)$$

$$\frac{\partial}{\partial t} M_X(t) = n\,p\,e^t\,(q + pe^t)^{n-1} \rightarrow (2)$$

We know that , $Mean = \mu_1' = \left.\frac{\partial}{\partial t} M_X(t)\right]_{t=0}$

$$= n\,p\,e^0\,(q + pe^0)^{n-1}$$

$$= n\,p\,(1)(q + p)^{n-1}$$

$$= n\,p\,(1)^{n-1}$$

$$\mu_1' = np$$

$\therefore$ Mean of a Binomial distribution is $\mu_1' = np$

Partially differentiating equation (2) with respect to 't'

$$\frac{\partial^2}{\partial t^2} M_X(t) = \frac{\partial}{\partial t}\left\{n\,p\,e^t\,(q + pe^t)^{n-1}\right\}$$

$$= n\,p\frac{\partial}{\partial t}\left[e^t\,(q + pe^t)^{n-1}\right]$$

$$= np\left\{e^t.\left(q+pe^t\right)^{n-1} + e^t(n-1)\left(q+pe^t\right)^{n-2}\left(pe^t\right)\right\}$$

$$\frac{\partial^2}{\partial t^2}M_X(t) = np\left\{e^t.\left(q+pe^t\right)^{n-1} + (n-1)p\,e^{2t}\left(q+pe^t\right)^{n-2}\right\} \rightarrow (3)$$

We know that , $\mu_2' = \left.\frac{\partial^2}{\partial t^2}M_X(t)\right]_{t=0}$

$$\mu_2' = np\left\{e^0.\left(q+pe^0\right)^{n-1} + (n-1)p\,e^{2(0)}\left(q+pe^0\right)^{n-2}\right\}$$

$$\mu_2' = np\left\{1.(q+p)^{n-1} + (n-1)p\,.1.(q+p)^{n-2}\right\}$$

$$\mu_2' = np\left\{(1)^{n-1} + (n-1)p(1)^{n-2}\right\}$$

$$\mu_2' = np\{1+(n-1)p\} = np\{1+np-p\}$$

$$\mu_2' = n^2p^2 - np^2 + np$$

By the definition of variance $\mu_2 = \mu_2' - (\mu_1')^2$

$$\mu_2 = \left[n^2p^2 - np^2 + np\right] - (np)^2$$

$$\mu_2 = \cancel{n^2p^2} - np^2 + np - \cancel{n^2p^2}$$

$$\mu_2 = np(1-p) = npq$$

Variance of a Binomial distribution is $\mu_2 = npq$

Probability generating function of a Binomial distribution :

A random variable 'x' is said to follow Binomial distribution with parameter 'n' and 'p' if it assumes only non- negative values and its probability mass function is given by

$$P(X=x) = \begin{cases} {}^nC_x p^x q^{n-x} & ;\ x = 0,1,2,3,.....n \\ 0 & ;\text{Otherwise} \end{cases}$$

By the definition of probability generating function, probability generating function of a Binomial distribution is

$$P_X(S) = E\left[S^x\right]$$

$$= \sum_x S^x P(X=x)$$

$$= \sum_{x=0}^{n} S^x\ {}^nC_x\ p^x\ q^{n-x}$$

$$= \sum_{x=0}^{n} {}^nC_x\ (S)^x\ \ p^x\ q^{n-x}$$

$$= \sum_{x=0}^{n} {}^nC_x\ (pS)^x\ q^{n-x}$$

$$={}^nC_0(pS)^0q^{n-0}+{}^nC_1(pS)^1q^{n-1}+{}^nC_2(pS)^2q^{n-2}+{}^nC_3(pS)^3q^{n-3}+.....+{}^nC_n(pS)^nq^{n-n}$$

$$\boxed{(x+y)^n={}^nC_0x^0y^n+{}^nC_1x^1y^{n-1}+{}^nC_2x^2y^{n-2}+{}^nC_3x^3y^{n-3}+{}^nC_4x^4y^{n-4}+.....+{}^nC_nx^ny^0}$$

$$P_X(S)=(pS+q)^n$$

$$P_X(S)=(q+pS)^n$$

$\therefore$ Probability generating function of a Binomial distribution is $P_X(S)=(q+pS)^n$

Recurrence relation for the probabilities of a binomial distribution :

Let 'x' be a discrete random variable which follows binomial distribution. Then its probability mass function is given by $P(X=x)={}^nC_xp^xq^{n-x}\rightarrow(1)\quad x=0,1,2,3,.....n$

If 'x' is a discrete random variable which follows binomial distribution, then $x+1$ is also a discrete random variable follows binomial distribution.

$$P(X+1)={}^nC_{x+1}p^{x+1}q^{n-(x+1)}\rightarrow(2)\quad x=0,1,2,3,.....n$$

From equation (1) and equation (2)

$$\frac{P(X+1)}{P(X)}=\frac{{}^nC_{x+1}p^{x+1}q^{n-(x+1)}}{{}^nC_xp^xq^{n-x}}$$

$$=\frac{\dfrac{\cancel{n!}}{(x+1)![n-(x+1)]!}\cancel{p^x}\,p\,\cancel{q^{n-x-1}}}{\dfrac{\cancel{n!}}{x![n-x]!}\cancel{p^x}\,\cancel{q^{n-x-1}}q}=\frac{x!(n-x)!}{(x+1)![n-x-1]!}\frac{p}{q}$$

$$\frac{P(X+1)}{P(X)}=\frac{\cancel{x!}(n-x)\cancel{[n-x-1]!}}{(x+1)\cancel{x!}\cancel{[n-x-1]!}}\frac{p}{q}$$

$$\frac{P(X+1)}{P(X)}=\frac{(n-x)}{(x+1)}\frac{p}{q}$$

$$P(X+1)=\left(\frac{n-x}{x+1}\right)\frac{p}{q}P(X)$$

Multiplying both sides by the total frequency 'N' then we get the recurrence relation for the expected frequencies of a binomial distribution.

$$N.P(X+1)=\left(\frac{n-x}{x+1}\right)\frac{p}{q}N.P(X)$$

$$E(X+1)=\left(\frac{n-x}{x+1}\right)\frac{p}{q}E(X)$$

Fitting of Binomial distribution :

Let , $x_1, x_2, x_3, \ldots\ldots\ldots\ldots x_n$ be 'n' values of a random variable 'x' with the corresponding frequencies .

$\therefore$ Mean of the random variable 'x` is given by $Mean(\bar{x}) = \dfrac{\sum fx}{N}$

Variance $= \dfrac{\sum f(x-\bar{x})^2}{N} = \dfrac{\sum fx^2}{N} - \left(\dfrac{\sum fx}{N}\right)^2$

If 'x' is a discrete random variable which follows binomial distribution, then by the properties of binomial distribution mean and variance are

Mean $(\bar{x}) = np \rightarrow (1)$ and variance $V(x) = npq \rightarrow (2)$

From equation (1) and Equation (2)

$$\frac{Variance}{Mean} = \frac{\cancel{np}q}{\cancel{np}} = \hat{q}$$

$$\hat{p} = 1 - \hat{q}$$

By substituting the value of $\hat{p}$ in equation (1) then we get the value of 'n'

The fitted binomial distribution for the given data is

$$P(X = x) = {}^nC_x \hat{p}^x \hat{q}^{n-x} \rightarrow (3) \quad x = 0,1,2,3,4,\ldots\ldots n$$

By substituting $x = 0,1,2,3,4,\ldots\ldots n$ in equation (3) then we get the corresponding probabilities of 'x'.

By multiplying each individual probability with the total frequency 'N' then we get the estimated frequencies for the given data.

Problem 1 : Fit a Binomial distribution by direct method from the following data and also estimate the expected frequencies.

x	0	1	2	3	4	5	6
f	7	64	140	210	132	75	12

Solution : Let the probability mass function of a binomial distribution be

$$P(X = x) = {}^nC_x p^x q^{n-x} \rightarrow (1) \quad x = 0,1,2,3,4,\ldots\ldots n$$

Here, $n = 6$

From the given data

x	f	fx
0	7	0
1	64	64
2	140	280
3	210	630
4	132	528
5	75	375
6	12	72
	640	1949

$\therefore$ Mean $\bar{x} = \frac{\sum fx}{N} = \frac{1949}{640} = 3.0453$

We know that, the mean of a binomial distribution is 'np'

$$np = 3.0453$$

$$6p = 3.0453$$

$$p = \frac{3.0453}{6} = 0.5076$$

$\hat{p} = 0.5076 \quad \hat{q} = 1 - \hat{p} = 1 - 0.5076 = 0.4924$

Substituting the values of n, p, q in equation (1) then we get

The fitted Binomial distribution for the given data is

$$P(X = x) = {}^6C_x (0.5076)^x (0.4924)^{6-x} \rightarrow (2) \quad x = 0,1,2,...6$$

Put $x = 0$ in equation (2) then $P(X = 0) = {}^6C_0 (0.5076)^0 (0.4924)^{6-0} = 0.0143$

Put $x = 1$ in equation (2) then $P(X = 1) = {}^6C_1 (0.5076)^1 (0.4924)^{6-1} = 0.0882$

Put $x = 2$ in equation (2) then $P(X = 2) = {}^6C_2 (0.5076)^2 (0.4924)^{6-2} = 0.2272$

Put $x = 3$ in equation (2) then $P(X = 3) = {}^6C_3 (0.5076)^3 (0.4924)^{6-3} = 0.3123$

Put $x = 4$ in equation (2) then $P(X = 4) = {}^6C_4 (0.5076)^4 (0.4924)^{6-4} = 0.2414$

Put $x = 5$ in equation (2) then $P(X = 5) = {}^6C_5 (0.5076)^5 (0.4924)^{6-5} = 0.0996$

Put $x = 6$ in equation (2) then $P(X = 0) = {}^6C_6 (0.5076)^6 (0.4924)^{6-6} = 0.0171$

Expected Frequencies :

$E(X = 0) = N.P(X = 0) = 640 \times 0.0143 = 9.1520 \cong 9$

$E(X = 1) = N.P(X = 1) = 640 \times 0.0882 = 56.4480 \cong 56$

$$E(X=2)=N.P(X=2)=640\times 0.2272=145.4080\cong 145$$

$$E(X=3)=N.P(X=3)=640\times 0.3123=199.8720\cong 200$$

$$E(X=4)=N.P(X=4)=640\times 0.2414=154.4960\cong 155$$

$$E(X=5)=N.P(X=5)=640\times 0.0996=63.7440\cong 64$$

$$E(X=6)=N.P(X=6)=640\times 0.0171=10.9440\cong 11$$

Estimated frequencies for the given data are

x	0	1	2	3	4	5	6
frequency	7	64	140	210	132	75	12
Expected frequency	9	56	145	200	155	64	11

Problem : 5 coins were tossed 1000 times and at each time the number of successes were noted and the following results were obtained.

No. of Successes	0	1	2	3	4	5
Frequency	38	144	342	287	164	25

Fit a binomial distribution and calculate the expected frequencies using direct method.

Solution : Let the equation of a binomial distribution is

$$P(X=x)={}^nC_x p^x q^{n-x} \rightarrow (1) \qquad x=0,1,2,3,.....,n$$

From the given data $n=5$

x	f	fx
0	38	0
1	144	144
2	342	684
3	287	861
4	164	656
5	25	125
	1000	2470

$\therefore$ Mean $\bar{x} = \frac{\sum fx}{N} = \frac{2470}{1000} = 2.47$

We know that, the mean of a binomial distribution is 'np'

$$np = 2.47$$
$$5p = 2.47$$
$$p = \frac{2.47}{5} = 0.4940$$

$\hat{p} = 0.4940 \quad \hat{q} = 1 - \hat{p} = 1 - 0.4940 = 0.5060$

Substituting the values of n, p, q in equation (1) then we get

The fitted Binomial distribution for the given data is

$$P(X = x) = {}^5C_x (0.4940)^x (0.5060)^{5-x} \rightarrow (2) \quad x = 0,1,2,...5$$

Put $x = 0$ in equation (2) then $P(X = 0) = {}^5C_0 (0.4940)^0 (0.5060)^{5-0} = 0.0332$

Put $x = 1$ in equation (2) then $P(X = 1) = {}^5C_1 (0.4940)^1 (0.5060)^{5-1} = 0.1619$

Put $x = 2$ in equation (2) then $P(X = 2) = {}^5C_2 (0.4940)^2 (0.5060)^{5-2} = 0.3162$

Put $x = 3$ in equation (2) then $P(X = 3) = {}^5C_3 (0.4940)^3 (0.5060)^{5-3} = 0.3087$

Put $x = 4$ in equation (2) then $P(X = 4) = {}^5C_4 (0.4940)^4 (0.5060)^{5-4} = 0.1507$

Put $x = 5$ in equation (2) then $P(X = 5) = {}^5C_5 (0.4940)^5 (0.5060)^{5-5} = 0.0294$

Expected Frequencies :

$E(X = 0) = N.P(X = 0) = 1000 \times 0.0332 = 33.20 \cong 33$

$E(X = 1) = N.P(X = 1) = 1000 \times 0.1619 = 161.9 \cong 162$

$E(X = 2) = N.P(X = 2) = 1000 \times 0.3162 = 316.2 \cong 316$

$E(X = 3) = N.P(X = 3) = 1000 \times 0.3087 = 308.7 \cong 309$

$E(X = 4) = N.P(X = 4) = 1000 \times 0.1507 = 150.7 \cong 151$

$E(X = 5) = N.P(X = 5) = 1000 \times 0.0294 = 29.4 \cong 29$

Estimated frequencies for the given data are

x	0	1	2	3	4	5
frequency	38	144	342	287	164	25
Expected frequency	33	162	316	309	151	29

Problem 3 : Fit a binomial distribution and calculate the expected frequencies by considering the total frequency as 200, if its mean and variance are 3.2 and 1.5 respectively.

Solution : Given that, mean and variance of a binomial distribution are 3.2 and 1.5 respectively.

By the properties of Binomial distribution, mean and variance of a Binomial distribution are 'np' and 'npq' respectively

$i.e.,$ Mean $np = 3.2 \rightarrow (1)$

Variance $npq = 1.5 \rightarrow (2)$

From equation (1) and equation (2)

$$\frac{\cancel{n}pq}{\cancel{n}p} = \frac{1.5}{3.2} = 0.4688$$

$$\hat{q} = 0.4688$$

$$\hat{p} = 1 - \hat{q} = 1 - 0.4688 = 0.5312$$

Substituting $\hat{p} = 0.5312$ in equation (1) then

$$n(0.5312) = 3.2$$

$$n = \frac{3.2}{0.5312} = 6.0241 \cong 6$$

The fitted Binomial distribution for the given data is

$$P(X = x) = {}^6C_x (0.5312)^x (0.4688)^{6-x} \rightarrow (3) \quad x = 0,1,2,...6$$

Put $x = 0$ in equation (2) then $P(X = 0) = {}^6C_0 (0.5312)^0 (0.4688)^{6-0} = 0.0106$

Put $x = 1$ in equation (2) then $P(X = 1) = {}^6C_1 (0.5312)^1 (0.4688)^{6-1} = 0.0722$

Put $x = 2$ in equation (2) then $P(X = 2) = {}^6C_2 (0.5312)^2 (0.4688)^{6-2} = 0.2044$

Put $x = 3$ in equation (2) then $P(X = 3) = {}^6C_3 (0.5312)^3 (0.4688)^{6-3} = 0.3089$

Put $x = 4$ in equation (2) then $P(X = 4) = {}^6C_4 (0.5312)^4 (0.4688)^{6-4} = 0.2625$

Put $x = 5$ in equation (2) then $P(X = 5) = {}^6C_5 (0.5312)^5 (0.4688)^{6-5} = 0.1190$

Put $x = 6$ in equation (2) then $P(X = 0) = {}^6C_6 (0.5312)^6 (0.4688)^{6-6} = 0.0225$

Expected Frequencies :

$$E(X = 0) = N.P(X = 0) = 200 \times 0.0106 = 2.12 \cong 2$$

$$E(X = 1) = N.P(X = 1) = 200 \times 0.0722 = 14.44 \cong 14$$

$$E(X = 2) = N.P(X = 2) = 200 \times 0.2044 = 40.88 \cong 41$$

$$E(X = 3) = N.P(X = 3) = 200 \times 0.3089 = 61.78 \cong 62$$

$$E(X = 4) = N.P(X = 4) = 200 \times 0.2625 = 52.5 \cong 53$$

$$E(X = 5) = N.P(X = 5) = 200 \times 0.1190 = 23.8 \cong 24$$

$E(X=6)=N.P(X=6)=200\times 0.0225=4.5\cong 4$

Estimated frequencies for the given data are

x	0	1	2	3	4	5	6
Expected frequency	2	14	41	62	53	24	4

Fitting of Binomial distribution using recurrence relation method:

Let 'x' be a discrete random variable which follows binomial distribution, then its probability mass function is given by $P(X=x)={}^nC_x p^x q^{n-x} \rightarrow (1)$; $x=0,1,2,3,.....n$

Let $x_1, x_2, x_3,.....x_n$ be 'n' values of the random variable 'x' with the corresponding frequencies $f_1, f_2, f_3,.....f_n$.

From the given data mean $\bar{x}=\dfrac{\sum fx}{N}$

By the properties of binomial distribution, mean of a random variable 'x' is 'np'

$$i.e., \quad np=\bar{x} \Rightarrow \hat{p}=\frac{\bar{x}}{n}$$

We know that $p+q=1 \Rightarrow \hat{q}=1-\hat{p}$

Substituting these estimates in equation (1) , then we get the fitted binomial distribution for the given data is

$P(X=x)={}^nC_x \hat{p}^x \hat{q}^{n-x} \rightarrow (2)$; $x=0,1,2,3,.....n$

Substituting $x=0$ in equation (2) then we get

$P(X=0)={}^nC_0 \hat{p}^0 \hat{q}^{n-0}=(1)(1)\hat{q}^n$

$P(X=0)=\hat{q}^n$

By the recurrence relation for the probabilities of binomial distribution

$$P(X=x+1)=\left(\frac{n-x}{x+1}\right)\left(\frac{\hat{p}}{\hat{q}}\right)P(X=x)\rightarrow (3)$$

Substituting $x=0,1,2,3,.....n-1$ in equation (3), then we get the individual probabilities for different values of 'x'.

By multiplying each individual probability with the total frequency 'N' then we get the expected frequencies for the given data.

Here the total expected frequency must be equal to the total observed frequency.

Problem : Fit a binomial distribution from the following data and estimate the expected frequencies by using recurrence relation method.

x	0	1	2	3	4	5	6	7	8
f	10	20	25	30	35	40	35	30	25

Solution : Let 'x' follows binomial distribution then its probability mass function is given by

$P(X=x)={}^nC_x p^x q^{n-x} \rightarrow (1)$; $x=0,1,2,3,.....n$

From the given data

x	f	fx
0	10	0
1	20	20
2	25	50
3	30	90
4	35	140
5	40	200
6	35	210
7	30	210
8	25	200
	250	1120

Mean $\bar{x} = \frac{\sum fx}{N} = \frac{1120}{250} = 4.48$

By the properties of binomial distribution

Mean of a binomial distribution is 'np'

$$i.e.,\ np = \bar{x} \Rightarrow \hat{p} = \frac{\bar{x}}{n} = \frac{4.48}{8} = 0.56$$

$\hat{q} = 1 - \hat{p} = 1 - 0.56 = 0.44$

Substituting $p = 0.56, q = 0.44$ and $n = 8$ in equation (1) then we get

The fitted binomial distribution for the given data is

$$P(X = x) = {}^8C_x (0.56)^x (0.44)^{8-x} \rightarrow (2) \quad x = 0,1,2,3,.....8$$

Put $x = 0$ in equation (1) then $P(X = 0) = {}^8C_0 (0.56)^0 (0.44)^{8-0} = 0.0014$

By the recurrence relation for the probabilities of binomial distribution

$$P(X = x+1) = \left(\frac{n-x}{x+1}\right)\left(\frac{\hat{p}}{\hat{q}}\right) P(X = x)$$

$$P(X = x+1) = \left(\frac{8-x}{x+1}\right)\left(\frac{0.56}{0.44}\right) P(X = x) \rightarrow (3)\ x = 0,1,2,3,.....7$$

Put $x = 0$ in equation (3) then

$$P(X = 0+1) = P(X = 1) = \left(\frac{8-0}{0+1}\right)\left(\frac{0.56}{0.44}\right) P(X = 0) = \left(\frac{8}{1}\right)\left(\frac{0.56}{0.44}\right) \times 0.0014 = 0.0143$$

Put $x = 1$ in equation (3) then

$$P(X = 1+1) = P(X = 2) = \left(\frac{8-1}{1+1}\right)\left(\frac{0.56}{0.44}\right) P(X = 1) = \left(\frac{7}{2}\right)\left(\frac{0.56}{0.44}\right) \times 0.0143 = 0.0637$$

Put $x=2$ in equation (3) then

$$P(X=2+1)=P(X=3)=\left(\frac{8-2}{2+1}\right)\left(\frac{0.56}{0.44}\right)P(X=2)=\left(\frac{6}{3}\right)\left(\frac{0.56}{0.44}\right)\times 0.0637=0.1622$$

Put $x=3$ in equation (3) then

$$P(X=3+1)=P(X=4)=\left(\frac{8-3}{3+1}\right)\left(\frac{0.56}{0.44}\right)P(X=3)=\left(\frac{5}{4}\right)\left(\frac{0.56}{0.44}\right)\times 0.1622=0.2580$$

Put $x=4$ in equation (3) then

$$P(X=4+1)=P(X=5)=\left(\frac{8-4}{4+1}\right)\left(\frac{0.56}{0.44}\right)P(X=4)=\left(\frac{4}{5}\right)\left(\frac{0.56}{0.44}\right)\times 0.2580=0.2627$$

Put $x=5$ in equation (3) then

$$P(X=5+1)=P(X=6)=\left(\frac{8-5}{5+1}\right)\left(\frac{0.56}{0.44}\right)P(X=5)=\left(\frac{3}{6}\right)\left(\frac{0.56}{0.44}\right)\times 0.2627=0.1672$$

Put $x=6$ in equation (3) then

$$P(X=6+1)=P(X=7)=\left(\frac{8-6}{6+1}\right)\left(\frac{0.56}{0.44}\right)P(X=6)=\left(\frac{2}{7}\right)\left(\frac{0.56}{0.44}\right)\times 0.1672=0.0608$$

Put $x=7$ in equation (3) then

$$P(X=7+1)=P(X=8)=\left(\frac{8-7}{7+1}\right)\left(\frac{0.56}{0.44}\right)P(X=7)=\left(\frac{1}{6}\right)\left(\frac{0.56}{0.44}\right)\times 0.0608=0.0097$$

Expected Frequencies :

$$E(X=0)=N\times P(X=0)=250\times 0.0014=0.35\cong 0$$
$$E(X=1)=N\times P(X=1)=250\times 0.0143=3.575\cong 4$$
$$E(X=2)=N\times P(X=2)=250\times 0.0637=15.925\cong 16$$
$$E(X=3)=N\times P(X=3)=250\times 0.1622=40.55\cong 41$$
$$E(X=4)=N\times P(X=4)=250\times 0.2580=64.5\cong 64$$
$$E(X=5)=N\times P(X=5)=250\times 0.2627=65.675\cong 66$$
$$E(X=6)=N\times P(X=6)=250\times 0.1672=41.8\cong 42$$
$$E(X=7)=N\times P(X=7)=250\times 0.0608=15.2\cong 15$$
$$E(X=8)=N\times P(X=8)=250\times 0.0097=2.425\cong 2$$

Properties of Binomial distribution :

- A random variable 'x' is said to follow Binomial distribution with parameter 'n' and 'p' if it assumes only non- negative values and its probability mass function is given by

$$P(X=x)=\begin{cases} {}^nC_x p^x q^{n-x} & ;\ x=0,1,2,3,.....n \\ 0 & ;\text{Otherwise} \end{cases}$$

- Mean of a binomial distribution is 'np'.
- Variance of a binomial distribution is 'npq'.
- Moment generating function of a Binomial distribution is $M_X(t)=\left(q+pe^t\right)^n$

- Characteristic function of a Binomial distribution is $\phi_X(t) = (q + pe^{it})^n$
- First four Raw moments of a binomial distribution are

  First Raw moment $\mu_1' = np$

  Second Raw moment $\mu_2' = n^2p^2 - np^2 + np$

  Third Raw moment $\mu_3' = np + 3n(n-1)p^2 + n(n-1)(n-2)p^3$

  Fourth Raw moment $\mu_4' = np + 7n(n-1)p^2 + 6n(n-1)(n-2)p^3 + n(n-1)(n-2)(n-3)p^4$
- First four central moments of a binomial distribution are

  First central moment $\mu_1 = 0$

  Second central moment $\mu_2 = npq$

  Third central moment $\mu_3 = npq(q-p)$

  Fourth central moment $\mu_4 = 3n^2p^2q^2 + npq(1-6pq)$
- Coefficient of skewness of a binomial distribution is $\gamma_1 = \frac{q-p}{\sqrt{npq}}$
- Coefficient of Kurtosis of a binomial distribution is $\gamma_2 = \frac{1-6pq}{npq}$
- Probability generating function of a binomial distribution is $P_X(S) = (q + pS)^n$
- First four cumulants of a binomial distribution are

  First cumulant $K_1 = np$ Second cumulant $K_2 = npq = \mu_2$

  Third cumulant $K_3 = \mu_3 = npq[q-p]$ Fourth cumulant $K_4 = \mu_4 - 3K_2^2$
- Recurrence relation for the probabilities of a binomial distribution is

  $$P(X = x+1) = \left(\frac{n-x}{x+1}\right)\left(\frac{p}{q}\right)P(X = x) \quad x = 0,1,2,3,.....n-1$$
- Recurrence relation for the expected frequencies of a binomial distribution is

  $$E(X = x+1) = \left(\frac{n-x}{x+1}\right)\left(\frac{p}{q}\right)E(X = x) \quad x = 0,1,2,3,.....n-1$$
- Recurrence relation for the moments of a binomial distribution is

  $$\mu_{r+1} = pq\left[nr\,\mu_{r-1} + \frac{\partial}{\partial p}\mu_r\right]$$

Poisson Distribution

Introduction :

Poisson distribution was introduced by a French mathematician Poisson. Binomial distribution tends to Poisson distribution when the number of trails 'n' is infinitely large.

Definition :

A discrete random variable 'x' is said to follow Poisson distribution then it's probability mass function is given by

$$P(X=x)=\begin{cases}\dfrac{e^{-\lambda}\lambda^{x}}{x!} & ;x=0,1,2,3,4,.....\\ 0 & ;Otherwise\end{cases}$$

Here 'λ' is the parameter.

Conditions of Poisson distribution :

1. Number of trails is very large.
2. Trails must be independent.
3. Probability of happening of more than one event in a very small interval of time is very small.

Mean and Variance of a Poisson distribution :

Let 'x' be a discrete random variable, which follows Poisson distribution then it's probability mass function is given by

$$P(X=x)=\begin{cases}\dfrac{e^{-\lambda}\lambda^{x}}{x!} & ;x=0,1,2,3,4,.....\\ 0 & ;Otherwise\end{cases}$$

Mean : By the definition of mathematical expectation, mathematical expectation of a random variable is equal to its arithmetic mean.

$$Mean\ \ E(x)=\sum_{x} x\,P(X=x)$$

$$=\sum_{x=0}^{\infty} x\frac{e^{-\lambda}\lambda^{x}}{x!}$$

$$=e^{-\lambda}\sum_{x=0}^{\infty}\cancel{x}\frac{\lambda\ \lambda^{x-1}}{\cancel{x}(x-1)!}$$

$$=\lambda e^{-\lambda}\sum_{x=1}^{\infty}\frac{\lambda^{x-1}}{(x-1)!}$$

$$=\lambda e^{-\lambda}\left\{\frac{\lambda^{1-1}}{(1-1)!}+\frac{\lambda^{2-1}}{(2-1)!}+\frac{\lambda^{3-1}}{(3-1)!}+\frac{\lambda^{4-1}}{(4-1)!}+.....\right\}$$

$$= \lambda e^{-\lambda}\left\{\frac{\lambda^0}{0!}+\frac{\lambda^1}{1!}+\frac{\lambda^2}{2!}+\frac{\lambda^3}{3!}+.....\right\}$$

$$= \lambda e^{-\lambda}\left\{1+\frac{\lambda}{1!}+\frac{\lambda^2}{2!}+\frac{\lambda^3}{3!}+.....\right\}$$

$$\boxed{e^x = 1+\frac{x}{1!}+\frac{x^2}{2!}+\frac{x^3}{3!}+.....}$$

$$Mean\ E(x) = \lambda\, \cancel{e^{-\lambda}}\left(\cancel{e^{\lambda}}\right)$$

$$Mean\ E(x) = \lambda$$

$\therefore$ Mean of a Poisson distribution is 'λ'

Variance :

Consider, $E(x^2) = \sum_x x^2\, P(X = x)$

$$E(x^2) = \sum_{x=0}^{\infty}(x^2 - x + x)P(X = x)$$

$$E(x^2) = \sum_{x=0}^{\infty}(x^2 - x)P(X = x) + \sum_{x=0}^{\infty} xP(X = x)$$

$$E(x^2) = \sum_{x=0}^{\infty}(x^2 - x)\frac{e^{-\lambda}\lambda^x}{x!} + \sum_{x=0}^{\infty} x\,\frac{e^{-\lambda}\lambda^x}{x!}$$

$$= \sum_{x=0}^{\infty}\cancel{x(x-1)}\,\frac{e^{-\lambda}\lambda^2\,\lambda^{x-2}}{\cancel{x(x-1)}(x-2)!} + \lambda$$

$$= \lambda^2 e^{-\lambda}\sum_{x=2}^{\infty}\frac{\lambda^{x-2}}{(x-2)!} + \lambda$$

$$= \lambda^2 e^{-\lambda}\left\{\frac{\lambda^{2-2}}{(2-2)!}+\frac{\lambda^{3-2}}{(3-2)!}+\frac{\lambda^{4-2}}{(4-2)!}+\frac{\lambda^{5-2}}{(5-2)!}+.....\right\} + \lambda$$

$$= \lambda^2 e^{-\lambda}\left\{\frac{\lambda^0}{0!}+\frac{\lambda^1}{1!}+\frac{\lambda^2}{2!}+\frac{\lambda^3}{3!}+.....\right\} + \lambda$$

$$= \lambda^2 e^{-\lambda}\left\{1+\frac{\lambda}{1!}+\frac{\lambda^2}{2!}+\frac{\lambda^3}{3!}+.....\right\} + \lambda$$

$$\boxed{e^x = 1+\frac{x}{1!}+\frac{x^2}{2!}+\frac{x^3}{3!}+.....}$$

$$E(x^2) = \lambda^2\cancel{e^{-\lambda}}\left\{\cancel{e^{\lambda}}\right\} + \lambda$$

$$E(x^2) = \lambda^2 + \lambda$$

By the definition of variance $V(x) = E(x^2) - [E(x)]^2$

$$V(x) = (\lambda^2 + \lambda) - (\lambda)^2$$

$$V(x) = \cancel{\lambda^2} + \lambda - \cancel{\lambda^2}$$

$$V(x) = \lambda$$

$\therefore$ Variance of a Poisson distribution is 'λ'

Moment generating function of a Poisson Distribution :

Let 'x' be a discrete random variable, which follows Poisson distribution then it's probability mass function is given by

$$P(X = x) = \begin{cases} \dfrac{e^{-\lambda}\lambda^x}{x!} & ; x = 0,1,2,3,4,..... \\ 0 & ; Otherwise \end{cases}$$

By the definition of moment generating function

$$M_X(t) = E\left[e^{tx}\right] = \sum_x e^{tx} P(X = x)$$

$$= \sum_{x=0}^{\infty} e^{tx} \frac{e^{-\lambda}\lambda^x}{x!}$$

$$= e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x \left(e^t\right)^x}{x!}$$

$$= e^{-\lambda} \sum_{x=0}^{\infty} \frac{\left(\lambda e^t\right)^x}{x!}$$

$$= e^{-\lambda} \left\{ \frac{\left(\lambda e^t\right)^0}{0!} + \frac{\left(\lambda e^t\right)^1}{1!} + \frac{\left(\lambda e^t\right)^2}{2!} + \frac{\left(\lambda e^t\right)^3}{3!} + ... \right\}$$

$$= e^{-\lambda} \left\{ 1 + \frac{\lambda e^t}{1!} + \frac{\left(\lambda e^t\right)^2}{2!} + \frac{\left(\lambda e^t\right)^3}{3!} + ... \right\}$$

$$\boxed{e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + ...}$$

$$M_X(t) = e^{-\lambda}\left[e^{\lambda e^t}\right] = e^{\lambda e^t - \lambda} = e^{\lambda\left(e^t - 1\right)}$$

$\therefore$ Moment generating function of Poisson distribution is $M_X(t) = e^{\lambda\left(e^t - 1\right)}$

Characteristic function of a Poisson Distribution :

Let 'x' be a discrete random variable, which follows Poisson distribution then it's probability mass function is given by

$$P(X = x) = \begin{cases} \dfrac{e^{-\lambda}\lambda^x}{x!} & ; x = 0,1,2,3,4,..... \\ 0 & ; Otherwise \end{cases}$$

By the definition of Characteristic function

$$\phi_X(t) = E\left[e^{itx}\right] = \sum_x e^{itx} P(X = x)$$

$$= \sum_{x=0}^{\infty} e^{itx} \frac{e^{-\lambda}\lambda^x}{x!}$$

$$= e^{-\lambda}\sum_{x=0}^{\infty}\frac{\lambda^x\left(e^{it}\right)^x}{x!}$$

$$= e^{-\lambda}\sum_{x=0}^{\infty}\frac{\left(\lambda e^{it}\right)^x}{x!}$$

$$= e^{-\lambda}\left\{\frac{\left(\lambda e^{it}\right)^0}{0!}+\frac{\left(\lambda e^{it}\right)^1}{1!}+\frac{\left(\lambda e^{it}\right)^2}{2!}+\frac{\left(\lambda e^{it}\right)^3}{3!}+.....\right\}$$

$$= e^{-\lambda}\left\{1+\frac{\lambda e^{it}}{1!}+\frac{\left(\lambda e^{it}\right)^2}{2!}+\frac{\left(\lambda e^{it}\right)^3}{3!}+.....\right\}$$

$$\boxed{e^x = 1+\frac{x}{1!}+\frac{x^2}{2!}+\frac{x^3}{3!}+.....}$$

$$\phi_X(t) = e^{-\lambda}\left[e^{\lambda e^{it}}\right] = e^{\lambda e^{it}-\lambda} = e^{\lambda\left(e^{it}-1\right)}$$

$\therefore$ Characteristic function of Poisson distribution is $\phi_X(t) = e^{\lambda\left(e^{it}-1\right)}$

Mean and variance of a Poisson Distribution using moment generating function :

Let 'x' be a discrete random variable which follows Poisson distribution, then its moment generating function is given by

$$M_X(t) = e^{\lambda\left(e^t-1\right)} \rightarrow (1)$$

Partially differentiating equation (1) with respect to 't' then

$$\frac{\partial}{\partial t}M_X(t) = \frac{\partial}{\partial t}e^{\lambda\left(e^t-1\right)}$$

$$= e^{\lambda\left(e^t-1\right)}\left[\lambda\left(e^t-0\right)\right]$$

$$= \lambda\, e^t\, e^{\lambda\left(e^t-1\right)}$$

$$\frac{\partial}{\partial t}M_X(t) = \lambda\ e^{\lambda\left(e^t-1\right)+t} \rightarrow (2)$$

We know that,

$$\mu_1' = \frac{\partial}{\partial t}M_X(t)\Big]_{t=0} = \lambda\ e^{\lambda\left(e^0-1\right)+0} = \lambda e^{\lambda(1-1)} = \lambda e^{\lambda(0)} = \lambda(1) = \lambda$$

Mean of a Poisson distribution is 'λ'

Partially differentiating equation (2) with respect to 't' then

$$\frac{\partial^2}{\partial t^2}M_X(t) = \lambda\frac{\partial}{\partial t}e^{\lambda\left(e^t-1\right)+t}$$

$$\frac{\partial^2}{\partial t^2}M_X(t) = \lambda\ e^{\lambda\left(e^t-1\right)+t}\left[\lambda\left(e^t-0\right)+1\right]$$

$$\frac{\partial^2}{\partial t^2}M_X(t) = \lambda\ e^{\lambda\left(e^t-1\right)+t}\left[\lambda e^t+1\right] \rightarrow (3)$$

We know that,

$$\mu_2' = \frac{\partial^2}{\partial t^2} M_X(t)\Big]_{t=0} = \lambda\ e^{\lambda(e^0-1)+0}\left[\lambda e^0 + 1\right]$$

$$= \lambda e^{\lambda(1-1)}\left[\lambda + 1\right]$$

$$= \lambda e^0\left[\lambda + 1\right]$$

$$= \lambda\left[\lambda + 1\right] = \lambda^2 + \lambda$$

Variance $V(x) = \mu_2' - (\mu_1')^2 = \left[\lambda^2 + \lambda\right] - (\lambda)^2 = \lambda^2 + \lambda - \cancel{\lambda^2} = \lambda$

Variance of a Poisson distribution is 'λ'

Moments of a Poisson distribution :

Let 'x' be a discrete random variable, which follows Poisson distribution then it's probability mass function is given by

$$P(X = x) = \begin{cases} \dfrac{e^{-\lambda}\lambda^x}{x!} & ; x = 0,1,2,3,4,\ldots\ldots \\ 0 & ; Otherwise \end{cases}$$

First raw moment $\mu_1' = \sum_x x\, P(X = x)$

$$= \sum_{x=0}^{\infty} x \frac{e^{-\lambda}\lambda^x}{x!}$$

$$= e^{-\lambda}\sum_{x=0}^{\infty} \cancel{x} \frac{\lambda\ \lambda^{x-1}}{\cancel{x}(x-1)!}$$

$$= \lambda e^{-\lambda}\sum_{x=1}^{\infty} \frac{\lambda^{x-1}}{(x-1)!}$$

$$= \lambda e^{-\lambda}\left\{\frac{\lambda^{1-1}}{(1-1)!} + \frac{\lambda^{2-1}}{(2-1)!} + \frac{\lambda^{3-1}}{(3-1)!} + \frac{\lambda^{4-1}}{(4-1)!} + \ldots\ldots\ldots\right\}$$

$$= \lambda e^{-\lambda}\left\{\frac{\lambda^0}{0!} + \frac{\lambda^1}{1!} + \frac{\lambda^2}{2!} + \frac{\lambda^3}{3!} + \ldots\ldots\ldots\right\}$$

$$= \lambda e^{-\lambda}\left\{1 + \frac{\lambda}{1!} + \frac{\lambda^2}{2!} + \frac{\lambda^3}{3!} + \ldots\ldots\ldots\right\}$$

$$\boxed{e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots\ldots\ldots}$$

$$\mu_1' = \lambda \cancel{e^{-\lambda}}\left(\cancel{e^{\lambda}}\right)$$

First raw moment $\boxed{\mu_1' = \lambda}$

Second raw moment , $\mu_2' = \sum_x x^2\ P(X = x)$

$$\mu_2' = \sum_{x=0}^{\infty}\left(x^2 - x + x\right)P(X = x)$$

$$\mu_2' = \sum_{x=0}^{\infty}\left(x^2 - x\right)P(X = x) + \sum_{x=0}^{\infty} xP(X = x)$$

$$\mu_2' = \sum_{x=0}^{\infty}\left(x^2 - x\right)\frac{e^{-\lambda}\lambda^x}{x!} + \sum_{x=0}^{\infty} x\,\frac{e^{-\lambda}\lambda^x}{x!}$$

$$= \sum_{x=0}^{\infty}\cancel{x(x-1)}\,\frac{e^{-\lambda}\lambda^2\,\lambda^{x-2}}{\cancel{x(x-1)}(x-2)!} + \lambda$$

$$= \lambda^2 e^{-\lambda}\sum_{x=2}^{\infty}\frac{\lambda^{x-2}}{(x-2)!} + \lambda$$

$$= \lambda^2 e^{-\lambda}\left\{\frac{\lambda^{2-2}}{(2-2)!} + \frac{\lambda^{3-2}}{(3-2)!} + \frac{\lambda^{4-2}}{(4-2)!} + \frac{\lambda^{5-2}}{(5-2)!} + \ldots\ldots\right\} + \lambda$$

$$= \lambda^2 e^{-\lambda}\left\{\frac{\lambda^0}{0!} + \frac{\lambda^1}{1!} + \frac{\lambda^2}{2!} + \frac{\lambda^3}{3!} + \ldots\ldots\right\} + \lambda$$

$$= \lambda^2 e^{-\lambda}\left\{1 + \frac{\lambda}{1!} + \frac{\lambda^2}{2!} + \frac{\lambda^3}{3!} + \ldots\ldots\right\} + \lambda$$

$$\boxed{e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots\ldots\ldots}$$

$$\mu_2' = \lambda^2 e^{-\lambda}\cancel{\left(e^{\lambda}\right)} + \lambda$$

$$\mu_2' = \lambda^2 + \lambda$$

Second raw moment of a Poisson distribution is $\mu_2' = \lambda^2 + \lambda$

Third raw moment $\mu_3' = \sum_x x^3\,P(X = x)$

$$\mu_3' = \sum_x\left[x(x-1)(x-2) + 3x(x-1) + x\right]P(X = x)$$

$$\mu_3' = \sum_x x(x-1)(x-2)P(X = x) + 3\sum_x x(x-1)P(X = x) + \sum_x x\,P(X = x)$$

$$\mu_3' = \sum_{x=0}^{\infty} x(x-1)(x-2)\frac{e^{-\lambda}\lambda^x}{x!} + 3\sum_{x=0}^{\infty} x(x-1)\frac{e^{-\lambda}\lambda^x}{x!} + \sum_{x=0}^{\infty} x\,\frac{e^{-\lambda}\lambda^x}{x!}$$

$$\mu_3' = e^{-\lambda}\sum_{x=0}^{\infty}\cancel{x(x-1)(x-2)}\frac{\lambda^3\;\lambda^{x-3}}{\cancel{x(x-1)(x-2)}(x-3)!} + 3\lambda^2 + \lambda$$

$$\mu_3' = \lambda^3 e^{-\lambda}\sum_{x=3}^{\infty}\frac{\lambda^{x-3}}{(x-3)!} + 3\lambda^2 + \lambda$$

$$\mu_3' = \lambda^3 e^{-\lambda}\left\{\frac{\lambda^{3-3}}{(3-3)!}+\frac{\lambda^{4-3}}{(4-3)!}+\frac{\lambda^{5-3}}{(5-3)!}+\frac{\lambda^{6-3}}{(6-3)!}+.....\right\}+3\lambda^2+\lambda$$

$$\mu_3' = \lambda^3 e^{-\lambda}\left\{\frac{\lambda^0}{0!}+\frac{\lambda^1}{1!}+\frac{\lambda^2}{2!}+\frac{\lambda^3}{3!}+.....\right\}+3\lambda^2+\lambda$$

$$\mu_3' = \lambda^3 e^{-\lambda}\left\{1+\frac{\lambda}{1!}+\frac{\lambda^2}{2!}+\frac{\lambda^3}{3!}+.....\right\}+3\lambda^2+\lambda$$

$$\mu_3' = \lambda^3 e^{-\lambda}\left\{e^{\lambda}\right\}+3\lambda^2+\lambda$$

$$\mu_3' = \lambda^3+3\lambda^2+\lambda$$

$\therefore$ Third raw moment of a Poisson distribution is $\mu_3' = \lambda^3+3\lambda^2+\lambda$

Fourth raw moment $\mu_4' = \sum_x x^4\, P(X=x)$

$$\mu_4' = \sum_x [x(x-1)(x-2)(x-3)+6x(x-1)(x-2)+7x(x-1)+x]P(X=x)$$

$$\mu_4' = \sum_x x(x-1)(x-2)(x-3)P(X=x)+6\sum_x x(x-1)(x-2)P(X=x)+\sum_x x(x-1)P(X=x)+\sum_x xP(X=x)$$

$$\mu_4' = \sum_{x=0}^{\infty} x(x-1)(x-2)(x-3)\frac{e^{-\lambda}\lambda^x}{x!}+6\sum_{x=0}^{\infty} x(x-1)(x-2)\frac{e^{-\lambda}\lambda^x}{x!}+7\sum_{x=0}^{\infty} x(x-1)\frac{e^{-\lambda}\lambda^x}{x!}+\sum_{x=0}^{\infty} x\frac{e^{-\lambda}\lambda^x}{x!}$$

$$\mu_4' = e^{-\lambda}\sum_{x=0}^{\infty} \cancel{x(x-1)(x-2)(x-3)}\frac{\lambda^4\ \lambda^{x-4}}{\cancel{x(x-1)(x-2)(x-3)(x-4)!}}+6\lambda^3+7\lambda^2+\lambda$$

$$\mu_4' = \lambda^4 e^{-\lambda}\sum_{x=4}^{\infty}\frac{\lambda^{x-4}}{(x-4)!}+6\lambda^3+7\lambda^2+\lambda$$

$$\mu_4' = \lambda^4 e^{-\lambda}\left\{\frac{\lambda^{4-4}}{(4-4)!}+\frac{\lambda^{5-4}}{(5-4)!}+\frac{\lambda^{6-4}}{(6-4)!}+\frac{\lambda^{7-4}}{(7-4)!}+.....\right\}6\lambda^3+7\lambda^2+\lambda$$

$$\mu_4' = \lambda^4 e^{-\lambda}\left\{\frac{\lambda^0}{0!}+\frac{\lambda^1}{1!}+\frac{\lambda^2}{2!}+\frac{\lambda^3}{3!}+.....\right\}+6\lambda^3+7\lambda^2+\lambda$$

$$\mu_4' = \lambda^4 e^{-\lambda}\left\{1+\frac{\lambda}{1!}+\frac{\lambda^2}{2!}+\frac{\lambda^3}{3!}+.....\right\}+6\lambda^3+7\lambda^2+\lambda$$

$$\mu_4' = \lambda^4 e^{-\lambda}\left\{e^{\lambda}\right\}+6\lambda^3+7\lambda^2+\lambda$$

$$\mu_4' = \lambda^4+6\lambda^3+7\lambda^2+\lambda$$

$\therefore$ Fourth raw moment of a Poisson distribution is $\mu_4' = \lambda^4+6\lambda^3+7\lambda^2+\lambda$

Central moments :

By the relationship between the central and raw moments,

First Central moment $\mu_1 = 0$

Second central moment $\mu_2 = \mu_2' - (\mu_1')^2 = (\lambda^2 + \lambda) - (\lambda)^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$

Third central moment $\mu_3 = \mu_3' - 3\mu_2'(\mu_1') + 2(\mu_1')^3$

$$\mu_3 = (\lambda^3 + 3\lambda^2 + \lambda) - 3(\lambda^2 + \lambda)(\lambda) + 2(\lambda)^3$$

$$\mu_3 = \lambda^3 + 3\lambda^2 + \lambda - 3\lambda^3 - 3\lambda^2 + 2\lambda^3$$

$$\mu_3 = \lambda$$

Fourth central moment $\mu_4 = \mu_4' - 4\mu_3'(\mu_1') + 6\mu_2'(\mu_1')^2 - 3(\mu_1')^4$

$$\mu_4 = (\lambda^4 + 6\lambda^3 + 7\lambda^2 + \lambda) - 4(\lambda^3 + 3\lambda^2 + \lambda)(\lambda) + 6(\lambda^2 + \lambda)(\lambda)^2 - 3(\lambda)^4$$

$$\mu_4 = \lambda^4 + 6\lambda^3 + 7\lambda^2 + \lambda - 4\lambda^4 - 12\lambda^3 - 4\lambda^2 + 6\lambda^4 + 6\lambda^3 - 3\lambda^4$$

$$\mu_4 = 3\lambda^2 + \lambda$$

First four central moments of a Poisson distribution are

1st Central moment $\mu_1 = 0$

2nd Central moment $\mu_2 = \lambda$

3rd Central moment $\mu_3 = \lambda$

4th Central moment $\mu_4 = 3\lambda^2 + \lambda$

Coefficient of skewness and Kurtosis of a Poisson distribution :

We know that $\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{\lambda^2}{\lambda^3} = \frac{1}{\lambda}$

Coefficient of skewness of a Poisson distribution is $\gamma_1 = \sqrt{\beta_1} = \sqrt{\frac{1}{\lambda}} = \frac{1}{\sqrt{\lambda}}$

We know that $\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{3\lambda^2 + \lambda}{\lambda^2} = \frac{3\lambda^2}{\lambda^2} + \frac{\lambda}{\lambda^2} = 3 + \frac{1}{\lambda}$

Coefficient of Kurtosis of a Poisson distribution is $\gamma_2 = \beta_2 - 3 = 3 + \frac{1}{\lambda} - 3 = \frac{1}{\lambda}$

Show that the Poisson distribution is a limiting case of Binomial Distribution :

Statement : If the number of trails 'n' becomes large, *i.e.*, $n \to \infty$ the probability of success 'p' becomes very small and $np = \lambda$ then the binomial distribution tends to Poisson distribution as a limiting case.

Proof : Let 'x' follows Binomial distribution, then it's probability mass function is given by

$$P(X=x)={}^{n}C_{x}\,p^{x}q^{n-x}$$

$$P(X=x)=\frac{n!}{x!(n-x)!}p^{x}q^{n-x}$$

$$=\frac{n(n-1)(n-2)(n-3).....(n-(x-3))(n-(x-2))(n-(x-1))\cancel{(n-x)!}}{x!\ \cancel{(n-x)!}}p^{x}q^{n-x}$$

$$=\frac{1}{x!}\{n(n-1)(n-2)(n-3).....(n-(x-3))(n-(x-2))(n-(x-1))\}p^{x}q^{n-x}$$

$$=\frac{1}{x!}\left\{n\left[n\left(1-\frac{1}{n}\right)\right]\left[n\left(1-\frac{2}{n}\right)\right]\left[n\left(1-\frac{3}{n}\right)\right].....\left[n\left(1-\frac{x-3}{n}\right)\right]\left[n\left(1-\frac{x-2}{n}\right)\right]\left[n\left(1-\frac{x-1}{n}\right)\right]\right\}p^{x}q^{n-x}$$

$$=\frac{n^{x}p^{x}}{x!}\left\{\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\left(1-\frac{3}{n}\right).....\left(1-\frac{x-3}{n}\right)\left(1-\frac{x-2}{n}\right)\left(1-\frac{x-1}{n}\right)\right\}q^{n-x}$$

Substituting $np=\lambda$ then $p=\frac{\lambda}{n}$

We know that, $q=1-p\Rightarrow q=1-\frac{\lambda}{n}$

$$P(X=x)=\frac{\lambda^{x}}{x!}\left\{\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\left(1-\frac{3}{n}\right).....\left(1-\frac{x-3}{n}\right)\left(1-\frac{x-2}{n}\right)\left(1-\frac{x-1}{n}\right)\right\}\left(1-\frac{\lambda}{n}\right)^{n-x}$$

Taking $\underset{n\to\infty}{lt}$ on both sides

$$\underset{n\to\infty}{lt}\,P(X=x)=\frac{\lambda^{x}}{x!}\underset{n\to\infty}{lt}\left\{\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\left(1-\frac{3}{n}\right).....\left(1-\frac{x-3}{n}\right)\left(1-\frac{x-2}{n}\right)\left(1-\frac{x-1}{n}\right)\right\}\underset{n\to\infty}{lt}\frac{\left(1-\frac{\lambda}{n}\right)^{n}}{\left(1-\frac{\lambda}{n}\right)^{x}}$$

$$\underset{n\to\infty}{lt}\,P(X=x)=\frac{\lambda^{x}}{x!}\left\{\left(1-\frac{1}{\infty}\right)\left(1-\frac{2}{\infty}\right)\left(1-\frac{3}{\infty}\right).....\left(1-\frac{x-3}{\infty}\right)\left(1-\frac{x-2}{\infty}\right)\left(1-\frac{x-1}{\infty}\right)\right\}\frac{1}{\left(1-\frac{\lambda}{\infty}\right)^{x}}\underset{n\to\infty}{lt}\left(1-\frac{\lambda}{n}\right)^{n}$$

$$\underset{n\to\infty}{lt}\,P(X=x)=\frac{\lambda^{x}}{x!}\{(1-0)(1-0)(1-0).....(1-0)(1-0)(1-0)\}\frac{1}{(1-0)^{x}}\underset{n\to\infty}{lt}\left(1-\frac{\lambda}{n}\right)^{n}$$

$$\underset{n\to\infty}{lt}\,P(X=x)=\frac{\lambda^{x}}{x!}\underset{n\to\infty}{lt}\left(1-\frac{\lambda}{n}\right)^{n}\to(1)$$

Consider, $\left(1-\frac{\lambda}{n}\right)^{n}={}^{n}C_{0}\left(\frac{\lambda}{n}\right)^{0}-{}^{n}C_{1}\left(\frac{\lambda}{n}\right)^{1}+{}^{n}C_{2}\left(\frac{\lambda}{n}\right)^{2}+{}^{n}C_{3}\left(\frac{\lambda}{n}\right)^{3}+.....+(-1)^{n}\,{}^{n}C_{n}\left(\frac{\lambda}{n}\right)^{n}$

$$=(1)(1)-n\left(\frac{\lambda}{n}\right)+\frac{n(n-1)}{2!}\frac{\lambda^{2}}{n^{2}}+\frac{n(n-1)(n-2)}{3!}\frac{\lambda^{3}}{n^{3}}-\frac{n(n-1)(n-2)(n-3)}{4!}\frac{\lambda^{4}}{n^{4}}+.....$$

$$=1-\lambda+\frac{\cancel{n^2}\left(1-\frac{1}{n}\right)}{2!}\frac{\lambda^2}{\cancel{n^2}}+\frac{\cancel{n^3}\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)}{3!}\frac{\lambda^3}{\cancel{n^3}}-\frac{\cancel{n^4}\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\left(1-\frac{3}{n}\right)}{4!}\frac{\lambda^4}{\cancel{n^4}}+.....$$

$$\left(1-\frac{\lambda}{n}\right)^n=1-\lambda+\left(1-\frac{1}{n}\right)\frac{\lambda^2}{2!}+\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\frac{\lambda^3}{3!}-\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\left(1-\frac{3}{n}\right)\frac{\lambda^4}{4!}+.....$$

Taking $\underset{n\to\infty}{lt}$ on both sides

$$\underset{n\to\infty}{lt}\left(1-\frac{\lambda}{n}\right)^n=\underset{n\to\infty}{lt}\left\{1-\lambda+\left(1-\frac{1}{n}\right)\frac{\lambda^2}{2!}+\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\frac{\lambda^3}{3!}-\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\left(1-\frac{3}{n}\right)\frac{\lambda^4}{4!}+.....\right\}$$

$$\underset{n\to\infty}{lt}\left(1-\frac{\lambda}{n}\right)^n=1-\lambda+\left(1-\frac{1}{\infty}\right)\frac{\lambda^2}{2!}+\left(1-\frac{1}{\infty}\right)\left(1-\frac{2}{\infty}\right)\frac{\lambda^3}{3!}-\left(1-\frac{1}{\infty}\right)\left(1-\frac{2}{\infty}\right)\left(1-\frac{3}{\infty}\right)\frac{\lambda^4}{4!}+.....$$

$$\underset{n\to\infty}{lt}\left(1-\frac{\lambda}{n}\right)^n=1-\lambda+(1-0)\frac{\lambda^2}{2!}+(1-0)(1-0)\frac{\lambda^3}{3!}-(1-0)(1-0)(1-0)\frac{\lambda^4}{4!}+.....$$

$$\underset{n\to\infty}{lt}\left(1-\frac{\lambda}{n}\right)^n=1-\frac{\lambda}{1!}+\frac{\lambda^2}{2!}+\frac{\lambda^3}{3!}-\frac{\lambda^4}{4!}+.....$$

$$\underset{n\to\infty}{lt}\left(1-\frac{\lambda}{n}\right)^n=e^{-\lambda}\to(2)$$

From equation (1) and equation (2)

$$\underset{n\to\infty}{lt}\ P(X=x)=\frac{\lambda^x}{x!}\left(e^{-\lambda}\right)$$

$$P(X=x)=\frac{e^{-\lambda}\lambda^x}{x!}\quad \text{where } x=0,1,2,3,4,.....$$

$\therefore$ Poisson distribution is a limiting case of Binomial Distribution.

Recurrence relation for the probabilities of Poisson distribution :

If 'x' is a discrete random variable which follows Poisson distribution then its probability mass function is given by

$$P(X=x)=\frac{e^{-\lambda}\lambda^x}{x!}\to(1)\quad \text{where } x=0,1,2,3,4,.....$$

If 'x' is a discrete random variable which follows Poisson distribution '$x+1$' is also a discrete random variable follows Poisson distribution.

$$P(X+1)=P(X=x+1)=\frac{e^{-\lambda}\lambda^{x+1}}{(x+1)!}\to(2)\quad \text{where } x=0,1,2,3,4,.....$$

From equation (1) and equation (2)

$$\frac{P(X+1)}{P(X)} = \frac{\dfrac{e^{-\lambda}\lambda^{x+1}}{(x+1)!}}{\dfrac{e^{-\lambda}\lambda^{x}}{x!}} = \frac{\cancel{e^{-\lambda}\lambda^{x}}\lambda}{(x+1)\cancel{x!}} \times \frac{\cancel{x!}}{\cancel{e^{-\lambda}\lambda^{x}}}$$

$$\frac{P(X+1)}{P(X)} = \frac{\lambda}{x+1}$$

$$P(X+1) = \frac{\lambda}{x+1}P(X) \qquad x = 0,1,2,3,4,\ldots\ldots\ldots\ldots$$

The above expression is known as the recurrence relation for the probabilities of a Poisson distribution.

Substituting $x = 0,1,2,3,4,\ldots\ldots\ldots\ldots$ in the above expression, then we get the corresponding probabilities.

Multiplying both sides by the total frequency 'N' then we get the recurrence relation for the expected frequencies of a Poisson distribution.

$$N.P(X+1) = \frac{\lambda}{x+1}N.P(X)$$

$$E(X+1) = \frac{\lambda}{x+1}E(X) \qquad x = 0,1,2,3,4,\ldots\ldots\ldots\ldots$$

The above expression is known as the recurrence relation for the expected frequencies of a Poisson distribution.

Substituting $x = 0,1,2,3,4,\ldots\ldots\ldots\ldots$ in the above expression, then we get the corresponding expected frequencies.

Fitting of Poisson distribution :

Let $x_1, x_2, x_3, \ldots\ldots x_n$ be 'n' values of a random variable 'x' with the corresponding frequencies $f_1, f_2, f_3, \ldots\ldots f_n$ then the mean of a random variable 'x' is given by

Mean $\bar{x} = \dfrac{\sum fx}{N}$

By the properties of Poisson distribution, mean of a random variable is given by 'λ'

$i.e.,\quad \bar{x} = \lambda$

$\therefore\ \hat{\lambda} = \bar{x}$

$\therefore$ The fitted Poisson distribution for the given data is

$P(X = x) = \dfrac{e^{-\hat{\lambda}}\hat{\lambda}^{x}}{x!}$ where $x = 0,1,2,3,4,\ldots\ldots\ldots$

Substituting $x = 0,1,2,3,4,.....$ in the above equation, then we get the corresponding probabilities.

Mode of a Poisson Distribution :

Mode is the value of a random variable which has the maximum probability or maximum frequency.

If 'x' is a discrete random variable which follows Poisson distribution then its probability mass function is given by

$$P(X = x) = \frac{e^{-\lambda}\lambda^{x}}{x!} \quad \text{where } x = 0,1,2,3,4,.....$$

If 'x' becomes mode, then 'x' has the maximum probability value $i.e.,\ P(x)$ is maximum.

Let $P(x-1)$ and $P(x+1)$ be the preceding and succeeding probabilities of $P(x)$ respectively.

- $P(x) \geq P(x-1)$
- $P(x) \geq P(x+1)$

Let $x-1$ be a discrete random variable which follows Poisson distribution, then its probability mass function is given by

$$P(x-1) = \frac{e^{-\lambda}\lambda^{x-1}}{(x-1)!} \quad \text{where } x = 0,1,2,3,4,.....$$

Similarly,

Let $x+1$ be a discrete random variable which follows Poisson distribution, then its probability mass function is given by

$$P(x+1) = \frac{e^{-\lambda}\lambda^{x+1}}{(x+1)!} \quad \text{where } x = 0,1,2,3,4,.....$$

i. Consider, $P(x) \geq P(x-1)$

$$\frac{e^{-\lambda}\lambda^{x}}{x!} \geq \frac{e^{-\lambda}\lambda^{x-1}}{(x-1)!}$$

$$\frac{\cancel{e^{-\lambda}\lambda^{x-1}}\lambda}{x\cancel{(x-1)!}} \geq \frac{\cancel{e^{-\lambda}\lambda^{x-1}}}{\cancel{(x-1)!}}$$

$$\frac{\lambda}{x} \geq 1 \Rightarrow \lambda \geq x$$

$$x \leq \lambda \rightarrow (1)$$

ii. Consider, $P(x) \geq P(x+1)$

$$\frac{e^{-\lambda}\lambda^{x}}{x!} \geq \frac{e^{-\lambda}\lambda^{x+1}}{(x+1)!}$$

$$\frac{\cancel{e^{-\lambda}}\lambda^{x}}{\cancel{x!}} \geq \frac{\cancel{e^{-\lambda}}\lambda^{x}\lambda}{(x+1)\cancel{x!}}$$

$$1 \geq \frac{\lambda}{x+1} \Rightarrow x+1 \geq \lambda \Rightarrow x \geq \lambda - 1$$

$$\lambda - 1 \leq x \rightarrow (2)$$

From equation (1) and equation (2)

$$\lambda - 1 \leq x \leq \lambda$$

Mode of a Poisson distribution is always lie between $\lambda - 1$ and λ.

Properties of Poisson distribution:

1. Let 'x' be a discrete random variable which follows a Poisson distribution then its probability mass function is given by

$$P(X = x) = \frac{e^{-\lambda}\lambda^{x}}{x!} \quad ; \; x = 0,1,2,3,4,\ldots\ldots\ldots$$

2. First four raw moments of a Poisson distribution are

   First Raw moment $\mu_1' = \lambda$

   Second Raw moment $\mu_2' = \lambda^2 + \lambda$

   Third Raw moment $\mu_3' = \lambda^3 + 3\lambda^2 + \lambda$

   Fourth Raw moment $\mu_4' = \lambda^4 + 6\lambda^3 + 7\lambda^2 + \lambda$

3. First four central moments of a Poisson distribution are

   First central moment $\mu_1 = 0$

   Second central moment $\mu_2 = \lambda$

   Third central moment $\mu_3 = \lambda$

   Fourth central moment $\mu_4 = 3\lambda^2 + \lambda$

4. Coefficient of skewness of a Poisson distribution is $\frac{1}{\sqrt{\lambda}}$

5. Coefficient of kurtosis of a Poisson distribution is $\frac{1}{\lambda}$
6. Mean of a Poisson distribution is λ
7. Variance of a Poisson distribution is λ
8. Moment generating function of a Poisson distribution is $M_X(t) = e^{\lambda[e^t - 1]}$
9. Characteristic function of a Poisson distribution is $\phi_X(t) = e^{\lambda[e^{it} - 1]}$
10. Probability generating function of a Poisson distribution is $P_X(S) = e^{\lambda(S-1)}$
11. First four cumulents of a Poisson distribution are

    First cumulent $K_1 = \lambda$

    Second cumulent $K_2 = \lambda$

    Third cumulent $K_3 = \lambda$

    Fourth cumulent $K_4 = \lambda$
12. Recurrence relation for the moments of a Poisson distribution are

$$\mu_{r+1} = \lambda\left[r\mu_{r-1} + \frac{\partial}{\partial\lambda}\mu_r\right]$$

13. Recurrence relation for the probabilities of a Poisson distribution are

$$P(X+1) = \frac{\lambda}{x+1}P(X)$$

14. Recurrence relation for the Expected frequencies of a Poisson distribution are

$$E(X+1) = \frac{\lambda}{x+1}E(X)$$

15. Poisson distribution is a limiting case of Binomial distribution.

1. A manufacturer known that 5% production of pins were defective. If they sells the pins in boxes of 100 guarantees not more than 10 pins will be defective. What is the approximate probability that box will fail to meet the guarantee.

Solution : Given that , number of pins in a box is $n = 100$

Also given that, probability of defective is 5 %

$$i.e., \quad p = 5\% = \frac{5}{100} = 0.05$$

By the properties of Poisson distribution $\lambda = np$

$$\lambda = 100 \times 0.05 = 5$$

The fitted Poisson distribution for the given data is

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!} = \frac{e^{-5}5^x}{x!} \quad \text{; where } x = 0,1,2,3,.....100$$

$$P(X > 10) = 1 - P(X \le 10)$$

$$= 1 - \sum_{x=0}^{10} \frac{e^{-5}5^x}{x!}$$

$$= 1 - e^{-5}\sum_{x=0}^{10} \frac{5^x}{x!}$$

2. In a Poisson distribution $P(X = 3) = \frac{2}{5}P(X = 2)$ then fit a Poisson distribution for the given data and also calculate

(i) $P(X \le 3)$

(ii) $P(2 < X \le 6)$

(iii) $P(X \ge 5)$

Solution : Given that, $P(X = 3) = \frac{2}{5}P(X = 2)$

$$\frac{e^{-\lambda}\lambda^3}{3!} = \frac{2}{5}\frac{e^{-\lambda}\lambda^2}{2!}$$

$$\frac{\cancel{e^{-\lambda}}\cancel{\lambda^3}}{3 \times \cancel{2!}} = \frac{2}{5}\frac{\cancel{e^{-\lambda}}\cancel{\lambda^2}}{\cancel{2!}}$$

$$\frac{\lambda}{3} = \frac{2}{5} \Rightarrow \lambda = \frac{6}{5} = 1.2$$

The fitted Poisson distribution for the given data is

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!} = \frac{e^{-1.2}(1.2)^x}{x!} \quad \text{; where } x = 0,1,2,3,.....$$

(i) $P(X \le 3) = \sum_{x=0}^{3} P(X = x)$

$$P(X \le 3) = \sum_{x=0}^{3} \frac{e^{-1.2}(1.2)^x}{x!} = e^{-1.2}\left[\frac{(1.2)^0}{0!} + \frac{(1.2)^1}{1!} + \frac{(1.2)^2}{2!} + \frac{(1.2)^3}{3!}\right]$$

$$= e^{-1.2}\left[1+1.2+\frac{1.44}{2}+\frac{1.728}{6}\right]$$

$$= e^{-1.2}\left[1+1.2+0.72+0.288\right]$$

$$= e^{-1.2}(3.208) = 0.9662$$

(ii) $P(2 < X \le 6) = P(X=3)+P(X=4)+P(X=5)+P(X=6)$

$$P(2 < X \le 6) = \frac{e^{-1.2}(1.2)^3}{3!}+\frac{e^{-1.2}(1.2)^4}{4!}+\frac{e^{-1.2}(1.2)^5}{5!}+\frac{e^{-1.2}(1.2)^6}{6!}$$

$$P(2 < X \le 6) = e^{-1.2}\left\{\frac{(1.2)^3}{6}+\frac{(1.2)^4}{24}+\frac{(1.2)^5}{120}+\frac{(1.2)^6}{720}\right\}$$

$$P(2 < X \le 6) = e^{-1.2}\{0.288+0.0864+0.0207+0.0041\}$$

$$P(2 < X \le 6) = 0.1201$$

(iii) $P(X \ge 5) = 1 - P(X \le 5)$

$$P(X \ge 5) = 1-\{P(X=0)+P(X=1)+P(X=2)+P(X=3)+P(X=4)+P(X=5)\}$$

$$P(X \ge 5) = 1-\left\{\frac{e^{-1.2}(1.2)^0}{0!}+\frac{e^{-1.2}(1.2)^1}{1!}+\frac{e^{-1.2}(1.2)^2}{2!}+\frac{e^{-1.2}(1.2)^3}{3!}+\frac{e^{-1.2}(1.2)^4}{4!}+\frac{e^{-1.2}(1.2)^5}{5!}\right\}$$

$$P(X \ge 5) = 1-e^{-1.2}\left\{\frac{(1.2)^0}{0!}+\frac{(1.2)^1}{1!}+\frac{(1.2)^2}{2!}+\frac{(1.2)^3}{3!}+\frac{(1.2)^4}{4!}+\frac{(1.2)^5}{5!}\right\}$$

$$P(X \ge 5) = 1-e^{-1.2}\{1+1.2+0.72+0.288+0.0864+0.0207\}$$

$$P(X \ge 5) = 1-e^{-1.2}\{3.2944\} = 1-0.9923 = 0.0077$$

3. Fit a Poisson distribution from the following data and estimate the expected frequencies by direct method.

x	0	1	2	3	4	5	6	7	8
f	71	112	117	57	27	11	3	1	1

Solution : From the given data

x	f	fx
0	71	0
1	112	112
2	117	234
3	57	171
4	27	108
5	11	55
6	3	18
7	1	7
8	1	8
	400	713

Mean $\bar{x} = \frac{\sum fx}{N} = \frac{713}{400} = 1.7825$

By the properties of Poisson distribution, $mean = \lambda$

$$\lambda = 1.7825$$

The fitted Poisson distribution for the given data is

$$P(X = x) = \frac{e^{-1.7825}(1.7825)^x}{x!} \rightarrow (1) \quad \text{Where } x = 0,1,2,3,.....$$

Put $x = 0$ then $P(X = 0) = \frac{e^{-1.7825}(1.7825)^0}{0!} = 0.1682$

Put $x = 1$ then $P(X = 1) = \frac{e^{-1.7825}(1.7825)^1}{1!} = 0.2998$

Put $x = 2$ then $P(X = 2) = \frac{e^{-1.7825}(1.7825)^2}{2!} = 0.2672$

Put $x = 3$ then $P(X = 3) = \frac{e^{-1.7825}(1.7825)^3}{3!} = 0.1588$

Put $x = 4$ then $P(X = 4) = \frac{e^{-1.7825}(1.7825)^4}{4!} = 0.0708$

Put $x = 5$ then $P(X = 5) = \frac{e^{-1.7825}(1.7825)^5}{5!} = 0.0252$

Put $x = 6$ then $P(X = 6) = \frac{e^{-1.7825}(1.7825)^6}{6!} = 0.0075$

Put $x = 7$ then $P(X=7) = \frac{e^{-1.7825}(1.7825)^7}{7!} = 0.0019$

Put $x = 8$ then $P(X=8) = \frac{e^{-1.7825}(1.7825)^8}{8!} = 0.0004$

Expected Frequencies :

$E(X=0) = N \times P(X=0) = 400 \times 0.1682 = 67.28 \cong 67$

$E(X=1) = N \times P(X=1) = 400 \times 0.2998 = 119.9 \cong 120$

$E(X=2) = N \times P(X=2) = 400 \times 0.2672 = 38.9 \cong 39$

$E(X=3) = N \times P(X=3) = 400 \times 0.1588 = 63.52 \cong 64$

$E(X=4) = N \times P(X=4) = 400 \times 0.0708 = 28.33 \cong 28$

$E(X=5) = N \times P(X=5) = 400 \times 0.0252 = 10.08 \cong 64$

$E(X=6) = N \times P(X=6) = 400 \times 0.0075 = 3 \cong 3$

$E(X=7) = N \times P(X=7) = 400 \times 0.0019 = 0.76 \cong 1$

$E(X=8) = N \times P(X=8) = 400 \times 0.0004 = 0.16 \cong 0$

Problem 4 : Fit a Poisson distribution for the given data and calculate the expected frequencies by using recurrence relation method.

x	0	1	2	3	4	5	6	7	8
f	56	156	132	92	37	22	4	0	1

Solution : from the given data

x	f	fx
0	56	0
1	156	156
2	132	264
3	92	276
4	37	148
5	22	110
6	4	24
7	0	0
8	1	8
	500	986

Mean $\bar{x} = \frac{\sum fx}{N} = \frac{986}{500} = 1.972$

By the properties of Poisson distribution, $mean = \lambda$

$$\lambda = 1.972$$

The fitted Poisson distribution for the given data is

$P(X = x) = \frac{e^{-1.972}(1.972)^x}{x!} \rightarrow (1)$ Where $x = 0,1,2,3,.....$

Put $x = 0$ then $P(X = 0) = \frac{e^{-1.972}(1.972)^0}{0!} = e^{-1.972} = 0.1392$

By the recurrence relation for the probabilities of Poisson distribution

$P(X+1) = \frac{\lambda}{x+1} P(X) \Rightarrow P(X+1) = \frac{1.972}{x+1} P(X) \rightarrow (2)$

Put $x = 0$ in equation (2) then $P(0+1) = P(1) = \frac{1.972}{0+1} P(0) = \frac{1.972}{0+1} \times 0.1392 = 0.2745$

Put $x = 1$ in equation (2) then $P(1+1) = P(2) = \frac{1.972}{1+1} P(1) = \frac{1.972}{2} \times 0.2745 = 0.2707$

Put $x = 2$ in equation (2) then $P(2+1) = P(3) = \frac{1.972}{2+1} P(2) = \frac{1.972}{3} \times 0.2707 = 0.1779$

Put $x = 3$ in equation (2) then $P(3+1) = P(4) = \frac{1.972}{3+1} P(3) = \frac{1.972}{4} \times 0.1780 = 0.0877$

Put $x = 4$ in equation (2) then $P(4+1) = P(5) = \frac{1.972}{4+1} P(4) = \frac{1.972}{5} \times 0.0877 = 0.0346$

Put $x = 5$ in equation (2) then $P(5+1) = P(6) = \frac{1.972}{5+1} P(5) = \frac{1.972}{6} \times 0.0346 = 0.0114$

Put $x = 6$ in equation (2) then $P(6+1) = P(7) = \frac{1.972}{6+1} P(6) = \frac{1.972}{7} \times 0.0114 = 0.0032$

Put $x = 7$ in equation (2) then $P(7+1) = P(8) = \frac{1.972}{7+1} P(7) = \frac{1.972}{8} \times 0.0032 = 0.0008$

Expected Frequencies :

$E(X = 0) = N \times P(X = 0) = 500 \times 0.1392 = 69.6 \cong 70$

$E(X = 1) = N \times P(X = 1) = 500 \times 0.2745 = 137.25 \cong 137$

$E(X = 2) = N \times P(X = 2) = 500 \times 0.2707 = 135.35 \cong 135$

$E(X = 3) = N \times P(X = 3) = 500 \times 0.1779 = 88.95 \cong 89$

$E(X = 4) = N \times P(X = 4) = 500 \times 0.0877 = 43.85 \cong 44$

$E(X = 5) = N \times P(X = 5) = 500 \times 0.0346 = 17.3 \cong 17$

$E(X = 6) = N \times P(X = 6) = 500 \times 0.0114 = 5.7 \cong 6$

$E(X = 7) = N \times P(X = 7) = 500 \times 0.0032 = 1.6 \cong 2$

$E(X = 8) = N \times P(X = 8) = 500 \times 0.0008 = 0.4 \cong 0$

Normal Distribution

Definition : A continuous random variable ‘ X ’ is said to follow Normal distribution if its probability density function is

$$f(x)=\begin{cases}\dfrac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} & ;Where\ -\infty<x<\infty,-\infty<\mu<\infty\\ 0 & ;Otherwise\end{cases}$$

Here $\sigma \geq 0$

If ‘ X ’ is a continuous random variable which follows Normal distribution then ‘ X ’ is called a normal variate and μ,σ^2 are the parameters of Normal distribution. A Normal variate can be represented as $X \sim N\left(\mu,\sigma^2\right)$

Mean and Variance of Normal distribution :

Mean :

Let ‘ X ’ be a continuous random variable which follows normal distribution then its mean is given by

$$Mean(\bar{x})=E(x)$$

$$=\int_x x\ f(x)\,dx$$

$$=\int_{-\infty}^{\infty} x\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}dx$$

$$=\int_{-\infty}^{\infty} x\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}dx$$

$$=\frac{1}{\sigma\sqrt{2\pi}}\int_{-\infty}^{\infty} xe^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}dx$$

Let $z=\dfrac{x-\mu}{\sigma} \Rightarrow x-\mu=z\sigma \Rightarrow x=\mu+z\sigma$

Differentiating on both sides

$$dx=0+dz\sigma \Rightarrow dx=\sigma\,dz$$

If $x=-\infty$ then $z=\dfrac{-\infty-\mu}{\sigma}=-\infty$

If $x=\infty$ then $z=\dfrac{\infty-\mu}{\sigma}=\infty$

$$E(x)=\frac{1}{\sigma\sqrt{2\pi}}\int_{-\infty}^{\infty}(\mu+z\sigma)e^{-\frac{1}{2}(z)^2}\sigma\,dz$$

$$E(x)=\frac{\cancel{\sigma}}{\cancel{\sigma}\sqrt{2\pi}}\left\{\int_{-\infty}^{\infty}\mu\ e^{-\frac{z^2}{2}}\,dz+\int_{-\infty}^{\infty}z\sigma\ e^{-\frac{z^2}{2}}\,dz\right\}$$

$$E(x)=\frac{1}{\sqrt{2\pi}}\left\{\mu\int_{-\infty}^{\infty}e^{-\frac{z^2}{2}}\,dz+\sigma\int_{-\infty}^{\infty}z\ e^{-\frac{z^2}{2}}\,dz\right\}$$

$$E(x)=\frac{1}{\sqrt{2\pi}}\left\{\mu\left(\sqrt{2\pi}\right)+\sigma(0)\right\}$$

$$E(x)=\frac{1}{\cancel{\sqrt{2\pi}}}\mu\left(\cancel{\sqrt{2\pi}}\right)$$

$$E(x)=\mu$$

$\therefore$ Mean of a Normal distribution is μ

Variance :

Let ‘ X ’ be a continuous random variable which follows normal distribution then its Variance is given by

$$Variance=E[x-\bar{x}]^2$$

$$=\int_{x}(x-\bar{x})^2\ f(x)\,dx$$

$$=\int_{-\infty}^{\infty}(x-\mu)^2\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}dx$$

$$=\frac{1}{\sigma\sqrt{2\pi}}\int_{-\infty}^{\infty}(x-\mu)^2e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}dx$$

Let $z=\dfrac{x-\mu}{\sigma}\ \Rightarrow\ x-\mu=z\sigma\ \Rightarrow\ x=\mu+z\sigma$

Differentiating on both sides

$$dx=0+dz\sigma\ \Rightarrow\ dx=\sigma\,dz$$

If $x=-\infty$ then $z=\dfrac{-\infty-\mu}{\sigma}=-\infty$

If $x=\infty$ then $z=\dfrac{\infty-\mu}{\sigma}=\infty$

$$Variance = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} (z\sigma)^2 e^{-\frac{1}{2}(z)^2} \sigma\, dz$$

$$Variance = \frac{\sigma^{\cancel{3}}}{\cancel{\sigma}\sqrt{2\pi}} \int_{-\infty}^{\infty} z^2 e^{-\frac{z^2}{2}}\, dz$$

$$Variance = \frac{\sigma^2}{\cancel{\sqrt{2\pi}}} \left(\cancel{\sqrt{2\pi}}\right)$$

$$Variance = \sigma^2$$

$\therefore$ Variance of a Normal distribution is σ^2

Moment generating function of Normal distribution :

Let 'X' be a continuous random variable which follows normal distribution then its moment generating function is given by

$$M_X(t) = E\left[e^{tx}\right]$$

$$= \int_x e^{tx} f(x)\, dx$$

$$= \int_{-\infty}^{\infty} e^{tx} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

$$= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tx} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

Let $z = \dfrac{x-\mu}{\sigma} \Rightarrow x - \mu = z\sigma \Rightarrow x = \mu + z\sigma$

Differentiating on both sides

$$dx = 0 + dz\sigma \Rightarrow dx = \sigma\, dz$$

If $x = -\infty$ then $z = \dfrac{-\infty - \mu}{\sigma} = -\infty$

If $x = \infty$ then $z = \dfrac{\infty - \mu}{\sigma} = \infty$

$$M_X(t) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{t(\mu + z\sigma)} e^{-\frac{1}{2}(z)^2} \sigma\, dz$$

$$M_X(t) = \frac{\cancel{\sigma}}{\cancel{\sigma}\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{t\mu} e^{tz\sigma} e^{-\frac{1}{2}(z)^2}\, dz$$

$$M_X(t)=\frac{e^{t\mu}}{\sqrt{2\pi}}\int_{-\infty}^{\infty}e^{tz\sigma-\frac{z^2}{2}}\,dz$$

$$M_X(t)=\frac{e^{t\mu}}{\sqrt{2\pi}}\int_{-\infty}^{\infty}e^{\frac{tz\sigma-z^2}{2}}\,dz$$

$$M_X(t)=\frac{e^{t\mu}}{\sqrt{2\pi}}\int_{-\infty}^{\infty}e^{-\frac{1}{2}\left(z^2-tz\sigma\right)}\,dz$$

$$M_X(t)=\frac{e^{t\mu}}{\sqrt{2\pi}}\int_{-\infty}^{\infty}e^{-\frac{1}{2}\left[\left(z^2-tz\sigma+(t\sigma)^2\right)-(t\sigma)^2\right]}\,dz$$

$$M_X(t)=\frac{e^{t\mu}}{\sqrt{2\pi}}\int_{-\infty}^{\infty}e^{-\frac{1}{2}(z-t\sigma)^2+\frac{t^2\sigma^2}{2}}\,dz$$

$$M_X(t)=\frac{e^{t\mu}}{\sqrt{2\pi}}\int_{-\infty}^{\infty}e^{-\frac{1}{2}(z-t\sigma)^2}\,e^{\frac{t^2\sigma^2}{2}}\,dz$$

$$M_X(t)=\frac{e^{t\mu+\frac{t^2\sigma^2}{2}}}{\sqrt{2\pi}}\int_{-\infty}^{\infty}e^{-\frac{1}{2}(z-t\sigma)^2}\,dz$$

$$M_X(t)=\frac{e^{t\mu+\frac{t^2\sigma^2}{2}}}{\cancel{\sqrt{2\pi}}}\left(\cancel{\sqrt{2\pi}}\right)$$

$$M_X(t)=e^{t\mu+\frac{t^2\sigma^2}{2}}$$

Moment generating function of a Normal distribution is $M_X(t)=e^{t\mu+\frac{t^2\sigma^2}{2}}$

Characteristic function of Normal distribution :

Let ‘ X ’ be a continuous random variable which follows normal distribution then its characteristic function is given by

$$\phi_X(t)=E\left[e^{itx}\right]$$

$$=\int_x e^{itx}f(x)\,dx$$

$$=\int_{-\infty}^{\infty}e^{itx}\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}\,dx$$

$$= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{itx} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

Let $z = \dfrac{x-\mu}{\sigma} \Rightarrow x-\mu = z\sigma \Rightarrow x = \mu + z\sigma$

Differentiating on both sides

$$dx = 0 + dz\sigma \Rightarrow \quad dx = \sigma\, dz$$

If $x = -\infty$ then $z = \dfrac{-\infty-\mu}{\sigma} = -\infty$

If $x = \infty$ then $z = \dfrac{\infty-\mu}{\sigma} = \infty$

$$\phi_X(t) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{it(\mu+z\sigma)} e^{-\frac{1}{2}(z)^2} \sigma\, dz$$

$$\phi_X(t) = \frac{\cancel{\sigma}}{\cancel{\sigma}\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{it\mu} e^{tz\sigma} e^{-\frac{1}{2}(z)^2} dz$$

$$\phi_X(t) = \frac{e^{it\mu}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{itz\sigma - \frac{z^2}{2}} dz$$

$$\phi_X(t) = \frac{e^{it\mu}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\frac{itz\sigma - z^2}{2}} dz$$

$$\phi_X(t) = \frac{e^{it\mu}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}\left(z^2 - itz\sigma\right)} dz$$

$$\phi_X(t) = \frac{e^{it\mu}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}\left[\left(z^2 - itz\sigma + (it\sigma)^2\right) - (it\sigma)^2\right]} dz$$

$$\phi_X(t) = \frac{e^{it\mu}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z-it\sigma)^2 - \frac{t^2\sigma^2}{2}} dz$$

$$\phi_X(t) = \frac{e^{it\mu}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z-it\sigma)^2} e^{-\frac{t^2\sigma^2}{2}} dz$$

$$\phi_X(t) = \frac{e^{it\mu - \frac{t^2\sigma^2}{2}}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z-it\sigma)^2} dz$$

$$\phi_X(t) = \frac{e^{it\mu - \frac{t^2\sigma^2}{2}}}{\cancel{\sqrt{2\pi}}}\left(\cancel{\sqrt{2\pi}}\right)$$

$$\phi_X(t) = e^{it\mu - \frac{t^2\sigma^2}{2}}$$

Characteristic function of a Normal distribution is $\phi_X(t) = e^{it\mu - \frac{t^2\sigma^2}{2}}$

Importance of Normal distribution :

Normal distribution has many number of applications in statistics, some are listed below.

1. The discrete probability distributions Binomial, Poisson, Hyper geometric etc., can be approximated to normal distribution for large values.
2. All small sample distributions Chi-square ,t, F and z approximated to normal distribution for large values.
3. The entire small, sample theory is based on the fundamental assumption that the parent population from which the sample is drawn is assumed to be normal.
4. In condition of large sample tests, the sampling distribution of sample means, sample variances, sample proportions etc., tends to normal distribution.
5. Entire large sample theory depends on the area property of normal distribution.
6. The theory of statistical quality control depends on 3σ limits developed by using $P(|z| \le 3) = 0.9973$ or $P(|z| > 3) = 0.0027$
7. Since error function follows normal distribution $\varepsilon \sim N(0, \sigma_e^2)$ it has a fundamental importance in the theory of errors in Astronomy.

Mean and Variance of Normal Distribution using moment generating function :

Let 'X' be a continuous random variable which follows normal distribution then its moment generating function is given by

$$M_X(t) = e^{t\mu + \frac{t^2\sigma^2}{2}} \rightarrow (1)$$

Partially differentiating equation (1) with respect to 't' then

$$\frac{\partial}{\partial t} M_X(t) = \frac{\partial}{\partial t} e^{t\mu + \frac{t^2\sigma^2}{2}}$$

$$\frac{\partial}{\partial t} M_X(t) = e^{t\mu + \frac{t^2\sigma^2}{2}}\left[1.\mu + \frac{2t\sigma^2}{2}\right]$$

$$\frac{\partial}{\partial t} M_X(t) = \left(\mu + t\sigma^2\right) e^{t\mu + \frac{t^2\sigma^2}{2}} \rightarrow (2)$$

We know that,

$$Mean\ \ E(x) = \frac{\partial}{\partial t} M_X(t)\Big]_{t=0} = \left(\mu + 0.\sigma^2\right) e^{0.\mu + \frac{0^2.\sigma^2}{2}} = (\mu + 0) e^0 = \mu.1 = \mu$$

$\therefore$ Mean of a normal distribution is μ.

Partially differentiating equation (2) with respect to 't' then

$$\frac{\partial^2}{\partial t^2} M_X(t) = \frac{\partial}{\partial t}\left[\left(\mu + t\sigma^2\right) e^{t\mu + \frac{t^2\sigma^2}{2}}\right]$$

$$\frac{\partial^2}{\partial t^2} M_X(t) = \left(0 + 1.\sigma^2\right) e^{t\mu + \frac{t^2\sigma^2}{2}} + \left(\mu + t\sigma^2\right)\left[\left(\mu + t\sigma^2\right) e^{t\mu + \frac{t^2\sigma^2}{2}}\right]$$

$$\frac{\partial^2}{\partial t^2} M_X(t) = \left[\left(\mu + t\sigma^2\right)^2 + \sigma^2\right] e^{t\mu + \frac{t^2\sigma^2}{2}} \rightarrow (3)$$

We know that

$$E\left(x^2\right) = \frac{\partial^2}{\partial t^2} M_X(t)\Big]_{t=0} = \left[\left(\mu + 0.\sigma^2\right)^2 + \sigma^2\right] e^{0.\mu + \frac{0^2.\sigma^2}{2}}$$

$$= \left[(\mu + 0)^2 + \sigma^2\right] e^0 = \left(\mu^2 + \sigma^2\right).1$$

$$= \mu^2 + \sigma^2$$

By the definition of variance

$$Variance = E\left[x^2\right] - \left[E(x)\right]^2$$

$$= \mu^2 + \sigma^2 - (\mu)^2$$

$$= \sigma^2$$

Variance of Normal distribution is σ^2

Moments of Normal distribution :

Let ‘ X ’ be a continuous random variable which follows normal distribution then its moment generating function is given by

$$M_X(t) = e^{t\mu + \frac{t^2\sigma^2}{2}} \rightarrow (1)$$

Partially differentiating equation (1) with respect to ‘ t ’ then

$$\frac{\partial}{\partial t} M_X(t) = \frac{\partial}{\partial t} e^{t\mu + \frac{t^2\sigma^2}{2}}$$

$$\frac{\partial}{\partial t} M_X(t) = e^{t\mu + \frac{t^2\sigma^2}{2}} \left[1.\mu + \frac{2t\sigma^2}{2} \right]$$

$$\frac{\partial}{\partial t} M_X(t) = \left(\mu + t\sigma^2\right) e^{t\mu + \frac{t^2\sigma^2}{2}} \rightarrow (2)$$

We know that,

$$\mu_1' = \frac{\partial}{\partial t} M_X(t) \Big]_{t=0} = \left(\mu + 0.\sigma^2\right) e^{0.\mu + \frac{0^2.\sigma^2}{2}} = \left(\mu + 0\right) e^0 = \mu.1 = \mu$$

$\therefore$ First Raw moment of a normal distribution is μ .

Partially differentiating equation (2) with respect to ‘ t ’ then

$$\frac{\partial^2}{\partial t^2} M_X(t) = \frac{\partial}{\partial t} \left[\left(\mu + t\sigma^2\right) e^{t\mu + \frac{t^2\sigma^2}{2}} \right]$$

$$\frac{\partial^2}{\partial t^2} M_X(t) = \left(0 + 1.\sigma^2\right) e^{t\mu + \frac{t^2\sigma^2}{2}} + \left(\mu + t\sigma^2\right) \left[\left(\mu + t\sigma^2\right) e^{t\mu + \frac{t^2\sigma^2}{2}} \right]$$

$$\frac{\partial^2}{\partial t^2} M_X(t) = \left[\left(\mu + t\sigma^2\right)^2 + \sigma^2 \right] e^{t\mu + \frac{t^2\sigma^2}{2}} \rightarrow (3)$$

We know that

$$\mu_2' = \frac{\partial^2}{\partial t^2} M_X(t) \Big]_{t=0} = \left[\left(\mu + 0.\sigma^2\right)^2 + \sigma^2 \right] e^{0.\mu + \frac{0^2.\sigma^2}{2}}$$

$$= \left[\left(\mu + 0\right)^2 + \sigma^2 \right] e^0 = \left(\mu^2 + \sigma^2\right).1$$

$$= \mu^2 + \sigma^2$$

Second Raw moment of a normal distribution is $\mu^2 + \sigma^2$

Partially differentiating equation (3) with respect to 't'

$$\frac{\partial^3}{\partial t^3} M_X(t) = \frac{\partial}{\partial t}\left\{ \left[\left(\mu + t\sigma^2\right)^2 + \sigma^2\right] e^{t\mu + \frac{t^2\sigma^2}{2}} \right\}$$

$$\frac{\partial^3}{\partial t^3} M_X(t) = \left[2\left(\mu + t\sigma^2\right)\left(0 + 1.\sigma^2\right) + 0\right] e^{t\mu + \frac{t^2\sigma^2}{2}} + \left[\left(\mu + t\sigma^2\right)^2 + \sigma^2\right]\left(\mu + t\sigma^2\right) e^{t\mu + \frac{t^2\sigma^2}{2}}$$

$$\frac{\partial^3}{\partial t^3} M_X(t) = \left\{2\sigma^2\left(\mu + t\sigma^2\right) + \sigma^2\left(\mu + t\sigma^2\right) + \left(\mu + t\sigma^2\right)^3\right\} e^{t\mu + \frac{t^2\sigma^2}{2}}$$

$$\frac{\partial^3}{\partial t^3} M_X(t) = \left\{3\sigma^2\left(\mu + t\sigma^2\right) + \left(\mu + t\sigma^2\right)^3\right\} e^{t\mu + \frac{t^2\sigma^2}{2}} \rightarrow (4)$$

We know that

$$\mu_3' = \left.\frac{\partial^3}{\partial t^3} M_X(t)\right]_{t=0} = \left[3\sigma^2\left(\mu + 0.\sigma^2\right) + \left(\mu + 0.\sigma^2\right)^3\right] e^{0.\mu + \frac{0^2.\sigma^2}{2}}$$

$$\mu_3' = \left[3\sigma^2(\mu) + (\mu)^3\right] e^0 = \left[\mu^3 + 3\mu\sigma^2\right] 1 = \mu^3 + 3\mu\sigma^2$$

Third Raw moment of a normal distribution is $\mu_3' = \mu^3 + 3\mu\sigma^2$

Partially differentiating equation (4) with respect to 't'

$$\frac{\partial^4}{\partial t^4} M_X(t) = \left[3\sigma^2\left(0 + 1.\sigma^2\right) + 3\left(\mu + t\sigma^2\right)^2\left(0 + 1.\sigma^2\right)\right] e^{t\mu + \frac{t^2\sigma^2}{2}} + \left\{3\sigma^2\left(\mu + t\sigma^2\right) + \left(\mu + t\sigma^2\right)^3\right\}\left(\mu + t\sigma^2\right) e^{t\mu + \frac{t^2\sigma^2}{2}}$$

$$\frac{\partial^4}{\partial t^4} M_X(t) = \left[3\sigma^4 + 3\sigma^2\left(\mu + t\sigma^2\right)^2 + 3\sigma^2\left(\mu + t\sigma^2\right)^2 + \left(\mu + t\sigma^2\right)^4\right] e^{t\mu + \frac{t^2\sigma^2}{2}}$$

$$\frac{\partial^4}{\partial t^4} M_X(t) = \left[3\sigma^4 + 6\sigma^2\left(\mu + t\sigma^2\right)^2 + \left(\mu + t\sigma^2\right)^4\right] e^{t\mu + \frac{t^2\sigma^2}{2}} \rightarrow (5)$$

We know that

$$\mu_4' = \left.\frac{\partial^4}{\partial t^4} M_X(t)\right]_{t=0} = \left[3\sigma^4 + 6\sigma^2\left(\mu + 0.\sigma^2\right)^2 + \left(\mu + 0.\sigma^2\right)^4\right] e^{0.\mu + \frac{0^2.\sigma^2}{2}}$$

$$\mu_4' = \left[3\sigma^4 + 6\sigma^2(\mu)^2 + (\mu)^4\right] e^0$$

$$\mu_4' = \left[3\sigma^4 + 6\mu^2\sigma^2 + \mu^4\right].1$$

$$\mu_4' = \mu^4 + 6\mu^2\sigma^2 + 3\sigma^4$$

Fourth raw moment of a normal distribution is $\mu_4' = \mu^4 + 6\mu^2\sigma^2 + 3\sigma^4$

Central moments :

By the relationship between central and raw moments

First Central moment $\mu_1 = 0$

Second Central moment $\mu_2 = \mu_2' - (\mu_1')^2 = \mu^2 + \sigma^2 - (\mu)^2 = \sigma^2$

Third Central moment $\mu_3 = \mu_3' - 3\mu_2'(\mu_1') + 2(\mu_1')^3$

$$\mu_3 = (\mu^3 + 3\mu\sigma^2) - 3(\mu^2 + \sigma^2)(\mu) + 2(\mu)^3 = \mu^3 + 3\mu\sigma^2 - 3\mu^3 - 3\mu\sigma^2 + 2\mu^3 = 0$$

Fourth Central moment $\mu_4 = \mu_4' - 4\mu_3'(\mu_1') + 6\mu_2'(\mu_1')^2 - 3(\mu_1')^4$

$$\mu_4 = (\mu^4 + 6\mu^2\sigma^2 + 3\sigma^4) - 4(\mu^3 + 3\mu\sigma^2)(\mu) + 6(\mu^2 + \sigma^2)(\mu)^2 - 3(\mu)^4$$

$$\mu_4 = \cancel{\mu^4} + \cancel{6\mu^2\sigma^2} + 3\sigma^4 - \cancel{4\mu^4} - \cancel{12\mu^2\sigma^2} + \cancel{6\mu^4} + \cancel{6\mu^2\sigma^2} - \cancel{3\mu^4}$$

$$\mu_4 = 3\sigma^4$$

Coefficient of Skewness and Coefficient of Kurtosis of a Normal distribution :

Coefficient of Skewness of a Normal distribution is

$$\gamma_1 = \sqrt{\beta_1} = \sqrt{\frac{\mu_3^2}{\mu_2^3}} = \sqrt{\frac{0^2}{(\sigma^2)^3}} = 0$$

Normal distribution is a symmetric distribution.

Coefficient of Kurtosis of a normal distribution is

$$\gamma_2 = \beta_2 - 3 = \frac{\mu_4}{\mu_2^2} - 3 = \frac{3\sigma^4}{(\sigma^3)^2} - 3 = \frac{3\sigma^4}{\sigma^4} - 3 = 3 - 3 = 0$$

Noramal distribution is a meso Kurtic distribution.